

Highlights

Hierarchical relations guide memory retrieval in sentence comprehension: Evidence from a local anaphor in Turkish

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- We tested the role of hierarchical *c-command* relations in memory retrieval.
- Three visual world studies examined antecedent retrieval for the Turkish reciprocal.
- Stimuli disentangled c-command from clause, case and subjecthood, and linear order.
- C-commanding targets were rapidly distinguished from non-c-commanding distractors.
- There was only limited evidence for interference from feature-matching distractors.
- Hierarchical relations between NPs determine their availability in early retrieval.

Hierarchical relations guide memory retrieval in sentence comprehension: Evidence from a local anaphor in Turkish

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Abstract

Real-time sentence comprehension relies on memory resources to establish long-distance syntactic dependencies. There is convincing evidence that formation of many dependencies is mediated by hierarchical constraints. However, it remains an open question how richly hierarchical information is represented and employed in memory processes. Here we ask whether hierarchical relations between noun phrases in a sentence as in *x c-commands y* (Reinhart, 1976) constrain antecedent retrieval for a local anaphor. We measure the real-time reactivation of c-commanding target antecedents and distractors in processing the Turkish reciprocal *birbirleri* via three visual world studies. Unlike existing studies that confounded multiple cues, we disentangled c-command from other structural information (clause-mateness, case, subjecthood) and linear order/recency. Experiment 1 compared the availability of c-commanding subjects and non-c-commanding, clause-mate, similarly case-marked distractors; and showed that c-commanding subjects were rapidly distinguished from distractors within the reciprocal window, irrespective of their linear order. Experiment 2 revealed that immediate availability of c-commanding subjects extends to c-commanding indirect objects. Experiment 3 was a pre-registered, high-power replication that yielded similar results. We found limited evidence for interference from distractors, which was not replicated. Overall, we find that hierarchical, item-to-item relations between noun phrases rapidly determine antecedent availability in retrieval beyond other cues. We suggest that hierarchical information may guide access to c-commanding items during retrieval if item representations include hierarchically informed features targeted by retrieval cues; alternatively, hierarchical information may shape the organization of items in memory, with c-commanding items represented in a privileged store allowing direct access during retrieval.

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1. Introduction

Hierarchical structure has long been considered the hallmark of human language syntax (Chomsky, 1957; Joshi et al., 1975; Kaplan and Bresnan, 1981; Steedman, 2000). Accordingly, it is thought to play a fundamental role in language comprehension by guiding the incremental interpretations that comprehenders assign to sentences, and has been the central focus of many theories of real-time language use (e.g., Dillon, 2011; Frazier and Clifton Jr., 1996; Gibson, 1998, 2000; Keshev et al., 2025; Kush, 2013; Levy, 2008; Phillips et al., 2011; Sturt, 2003; Wagers, 2008; cf. Frank and Bod, 2011).

One way in which hierarchical structure is important is in how it appears to constrain the interpretation of long-distance dependencies that span multiple intervening words. For example, in (1) the anaphor *each other* must be interpreted in relation to its antecedent *kids*. Despite the presence of several feature-matching (i.e., plural) referents in the sentence, i.e., *seniors* and *neighbors*, *kids* is the only available antecedent. This is due to its hierarchical position relative to the anaphor within the sentence’s syntax: *Kids* is local (e.g., clause-mate) to the anaphor and hierarchically prominent (e.g., in a c-commanding position) relative to it within the embedded clause (Chomsky, 1981; Reinhart, 1976), which are the two configurational constraints on what makes a possible antecedent for a local anaphor.

- (1) The seniors_i mentioned that the neighbors_j’ kids_k playfully teased *each other*_{*i/*j/k}.

Such hierarchical constraints are commonly defined in terms of structural relationships between pairs of linguistic items within a sentence, as in *x is local to y* or *x c-commands y*.

One question for theories of real-time dependency formation is how to integrate such relational constraints between items with theories of psychological mechanisms that subserve real-time dependency formation (Alcocer and Phillips, 2012; Dillon, 2011; Kush, 2013; Wagers, 2008). On the one hand, there is an abundance of evidence that comprehenders activate numerous, grammatically licit interpretations of linguistic input rapidly and in parallel (e.g., MacDonald et al., 1994). Such a view would suggest that any grammatically licensed interpretation of the input could be made available very early in the comprehension process.

On the other hand, algorithmic-level accounts of dependency formation often invoke a content-based, associative retrieval mechanism (e.g., Lewis and Vasishth,

2005; Lewis et al., 2006; McElree, 2000, 2006; Van Dyke and McElree, 2011). On this view, linguistic items are assumed to be encoded with their *item-specific* features in short term memory, using structural (e.g., subjecthood, case) and morphological (e.g., number, gender) information. Dependency formation then requires reactivating these encodings on demand in response to a set of retrieval cues that are checked against the feature content of individual items in memory. On this view, dependency formation is expected to be rapid when it can be cued by the features of a target item, as this will allow rapid retrieval of that item from working memory (Lewis and Vasishth, 2005; Lewis et al., 2006; McElree et al., 2003). But relational notions such as c-command or relative linear order are less straightforwardly encoded in such an architecture (Alcocer and Phillips, 2012; Dillon and Keshev, 2024; Kush, 2013; Wagers, 2008). If memory constraints require comprehenders to infer or reconstruct the relationship between two elements to establish that they are in an appropriate configurational relationship (Dillon, 2011; Dillon et al., 2014; Kush et al., 2015), then we might expect dependency formation to be slower or less reliable when such relational processing is needed (McElree and Doshier, 1993).

To address the time course and fidelity of such relational linguistic processing, an enormous amount of research has looked into the real-time processing of reflexives and other anaphors. This work has yielded compelling evidence that comprehenders rapidly employ structural constraints in linking anaphors with their antecedents (see citations in Appendix A). Still, it remains an open question whether this body of work definitively establishes that comprehenders use hierarchical constraints that necessarily refer to relational information between linguistic items in antecedent retrieval, or if it is possible to reconcile these findings with an item-based, feature-driven retrieval mechanism (Dillon et al., 2013; Kush and Phillips, 2014; Kush et al., 2015). As a result, the question of whether hierarchical relational information is available to guide dependency formation in early stages of comprehension remains contentious.

Here we aim to contribute to this literature by asking: Is the hierarchical information that is exploited in antecedent retrieval from memory in real-time constrained by hierarchical relations between individual noun phrases (NPs) in a sentence as in *x c-commands y*? We report three visual world studies that measure the moment-by-moment reactivation of potential antecedents in processing a local anaphor in Turkish. We employed carefully designed stimuli that isolated the c-command relation between NPs in a sentence while eliminating the role of other structural information such as clause, case and subjecthood, and linear order/recency. To preview our results, we find that the abstract hierarchical relation between NPs in a sentence determines antecedent availability above and beyond other sources of information from the earliest moments of interpretation of an anaphor. The finding that abstract

hierarchical relations are available to guide interpretation from the earliest moments of anaphor comprehension constrains psycholinguistic theories of dependency formation, which we take up in the discussion.

1.1. The role of hierarchical relations in real-time processing

The processing of anaphors such as reflexives (*himself, herself, themselves*) and reciprocals (*each other*) has been a focus of research into this area because they are common linguistic elements that are subject to relatively well understood syntactic constraints on interpretation, expressed in (for example) Principle A of Binding Theory (Chomsky, 1981). There are two critical hierarchical constraints that define potential antecedents of a local anaphor: The target antecedent must appear within a sufficiently local domain (e.g., the clause of the anaphor), and stand in a hierarchically available (e.g., c-commanding) position with respect to the anaphor. In addition, antecedents must agree with the features of the anaphor (i.e., in person, number and gender).

The *c-command* constraint is based on abstract relations defined on the basis of the hierarchical organization of individual noun phrases (NPs) in a sentence’s syntactic configuration. Formally, a node α c(onstituent)-commands a node β if β is α ’s sister or is contained within α ’s sister in the tree structure (Büring, 2005; Reinhart, 1976, 1983). For example, in the sentence *The seniors mentioned that the neighbors’ kids playfully teased each other* (repeated from (1) above), the target subject *kids* c-commands the anaphor *each other* because the anaphor is contained within the verb phrase (VP) that is the sister of the target subject’s NP, while *neighbors* does not c-command the anaphor as it is embedded within this subject’s NP as a possessor and is not a sister to the same VP¹.

Many empirical studies on the processing of anaphors have shown that comprehenders rapidly distinguish hierarchically available, e.g., c-commanding and clause-mate, targets from distractors, which is consistent with the use of relational information in early memory processes (Dillon et al., 2013, 2014; Kush et al., 2015; Sturt, 2003; Xiang et al., 2009). However, it remains unclear whether these findings

¹We adopt the notion of c-command to characterize the abstract geometric relations within constituent structure that constrain anaphoric dependencies. We acknowledge that in addition to c-command, a range of other proposals have been advanced in the literature to identify similar relational constraints within a configurational hierarchy including *command* (Langacker, 1969), *kommand* (Lasnik, 1976), *maximal-projection-command* (Chomsky, 1981) and *precede-and-command* (Bruening, 2014). Since all of these proposals converge on the core idea that a hierarchical relational constraint between an anaphor and its antecedent is necessary to describe anaphoric dependencies, in this paper we use the notion of c-command as it is widely recognized in syntactic theory.

necessarily implicate the use of an abstract hierarchical notion like c-command in determining antecedent availability, as individual studies typically admit multiple interpretations. For example, Dillon et al. (2014) argued that comprehenders use structural information of ‘clausemateness’ to constrain the initial stages of memory retrieval when processing the Mandarin Chinese anaphor *ziji* (see also Dillon et al. (2016)). While it is possible to consider the notion of ‘clausemate’ to define a relational constraint between two items in this way, this is not a logical necessity. One could instead argue that clause level information is encoded as an item-level feature, which may have an independent psychological motivation if a clause is considered as a single temporal context (Howard and Kahana, 2002) in the form of events anchored by its verb (Wagers, 2008). In that sense, identifying items within a clause would reflect their contextual organization in memory and would allow a simple, feature-based encoding of clausal context that can be cued in retrieval (Balachandran, 2024; Polyn et al., 2009; Sederberg et al., 2008).

More compellingly, several studies have explicitly tried to isolate the role of a hierarchical relation like c-command in processing bound variable pronouns (Cunnings et al., 2015; Kush et al., 2015; Moulton and Han, 2018). For example, Kush et al. (2015) compared the processing of bound variable and coreferential pronouns. They conducted an eye-tracking-while-reading experiment with stimuli as in (2) crossing antecedent type, i.e., quantificational (*any janitor*) or referential (*the janitor*), and the c-command relation between the pronoun *he* and the antecedent using either *when*- or *but*-clauses. In (2-a) the bound variable reading for *he* is available when it is in a *when*-clause, which is attached to the embedded verb phrase and c-commanded by *any janitor*. The bound variable reading is unavailable when *he* is in a *but*-clause, which is attached to the root of the sentence and not c-commanded by *any janitor*. Coreference interpretation between *he* and *the janitor* in (2-b), on the other hand, is available irrespective of the type of the clause. If antecedent retrieval for bound variable pronouns uses hierarchical relational information such as the c-command constraint, extra processing difficulty is expected only when the quantificational NP *any janitor* does not c-command the pronoun *he* in a *but*-clause.

- (2) a. Kathi didn’t think *any janitor* liked performing his custodial duties, when/but *he* had to clean up messes left after prom anyway.
- b. Kathi didn’t think *the janitor* liked performing his custodial duties, when/but *he* had to clean up messes left after prom anyway.

The results showed that while there was no evidence for a difference between the two conditions with a referential NP, there was an increased difficulty in reading the post-pronoun region when *any janitor* did not c-command *he* in the *but*-clause compared

to the *when*-clause condition. In a follow-up eye-tracking experiment they further demonstrated that feature match of quantificational NPs in a non-c-commanding position did not affect early reading measures, with only suggestive evidence for delayed interference effects from gender-matching, non-c-commanding quantificational NPs in later measures (see Cunnings et al. (2015) for similar findings). In a later study, Moulton and Han (2018) disentangled the syntactic notion of c-command from the semantic notion of scope, and found that it was the syntactic c-command relation that seemed to determine antecedent accessibility in processing bound variable pronouns. Studies like these suggest that initial stages of antecedent retrieval can use the hierarchical information of NPs in distinguishing the available and unavailable quantificational NPs in interpreting bound variable pronouns.

Still, it is possible that Kush et al.’s (and Cunnings et al.’s and Moulton and Han’s) findings are driven by relations among clauses, rather than fine-grained relations between NPs. In other words, in Kush et al.’s stimuli, the critical c-command constraint between the quantificational NP and the pronoun can also be established between the clauses in which they appear. The clause with the quantificational NP in (2-a) c-commands the *when*-clause and hence the pronoun that it embeds, while it does not c-command the *but*-clause nor the pronoun in it. Thus, although their findings do provide evidence for the use of hierarchical relations in memory retrieval, they cannot be unambiguously interpreted as evidence for the immediate use of the c-command relation between NPs.

Lastly, there is also empirical evidence that the hierarchical organization of NPs impacts the availability of their memory representations for further operations. For example, testing subject-verb agreement cases in a probe recognition task, Franck and Wagers (2020) showed that attractor nouns that c-command the verb are more readily available in memory than those that do not c-command it, such as *dafrans* in **Which dafrans of the brapou do you say that the bostron defend?* vs. **The brapou of which dafrans do you say that the bostron defend?*, respectively (translated from stimuli in French consisting nonce words) (see Franck et al., 2006, 2010, 2020, for similar findings in judgment tasks). Although such findings suggest that hierarchical structure has a role in memory representations, they do not necessarily indicate that c-commanding nouns are represented or accessed from memory on the basis of their hierarchical relation with the verb. Rather, as discussed by Franck and Wagers, these findings may be attributed to the increased activation levels of c-commanding nouns in memory due to their prominent hierarchical position.

1.2. A review of local anaphor processing: A gap on the role of c-command

To date, dozens of studies have explored the real-time processing of anaphors, potentially establishing them as one of the most commonly studied items in psycholinguistic research. On the surface, there appears to be a considerable amount of evidence that memory retrieval in real-time processing of anaphoric dependencies is sensitive to hierarchical information in accessing the target antecedent (e.g., Dillon et al., 2013; Nicol and Swinney, 1989; Sturt, 2003; Xiang et al., 2009). However, while these results provide clear evidence that comprehenders use some type of structural information in retrieval, they do not answer the question of whether an abstract relational notion like c-command *per se* is used to distinguish NPs within a clause.

Against the rich empirical background on anaphoric dependencies, it is perhaps surprising that the role of c-command in resolving anaphoric items remains unclear. But one key observation about existing studies is that most used stimuli that confound multiple sources of structural information in defining the target of retrieval, failing to uniquely isolate the role of c-command in retrieval. This leads to the possibility that a range of structural, item-level information may be used to retrieve the target antecedent instead of the relational notion of c-command. To substantiate this observation, we conducted a review of the literature on processing local anaphors, summarized in the table in Appendix A. Our review included 44 experiments from 27 crosslinguistic studies on local anaphors that included structurally unavailable distractors. The studies were predominantly in English, with some in Chinese, German, Hindi, Korean, Russian, Swedish, Tagalog and Turkish. We examined the experimental stimuli in these studies to determine what structural/relational information was used to distinguish the target and distractor. Our review also included details about the studies (e.g., language under investigation, number of participants) and summarized the findings as originally presented.

We found that no study in our review directly isolated the role of c-command *per se* without confounding it with other potential structural cues such as clause-mateness and/or grammatical case. Importantly, these are cues that are plausibly construed as item-level features. A summary of our review is illustrated with a Venn diagram in Figure 1. The diagram shows the structural/relational cues manipulated to distinguish the c-commanding, clause-mate, nominative-marked (or the relevant case marking in the language under investigation) target and the distractor. The circled areas labeled *C-command*, *Clause* and *Case* include studies that manipulated each of these features individually when creating a distractor, with their intersections listing those that manipulated their corresponding combination. The gray-shaded area highlights the absence of studies that manipulate only the c-command relation between NPs.

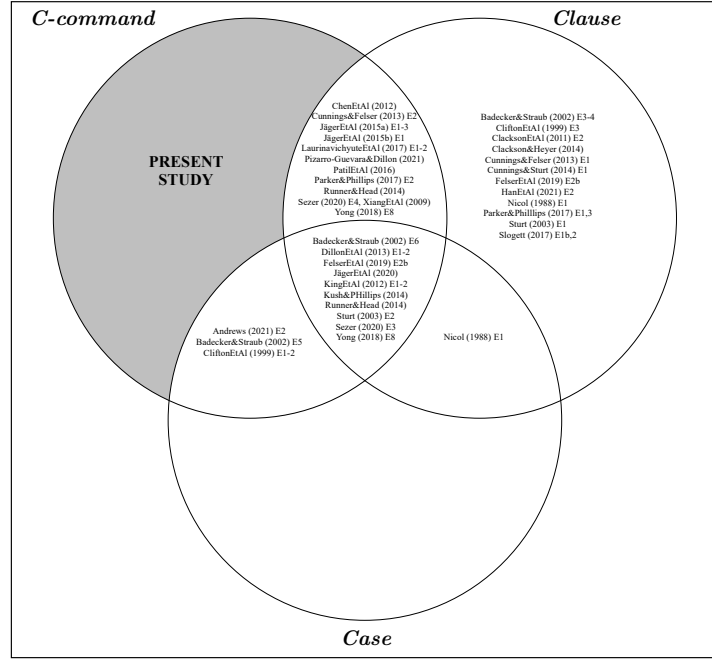


Figure 1: Venn diagram illustrating the structural/relational information manipulated in previous studies on local anaphor processing to distinguish structurally available targets from distractors.

To illustrate further, 12 experiments from 10 studies created a distractor by manipulating all three structural/relational features, using a non-c-commanding, non-clause-mate, differently-case-marked distractor as in (3-a). Fifteen experiments from 12 of the reviewed studies manipulated both c-command and clause features as in (3-b). Four experiments from 3 studies manipulated both c-command and case features as in (3-c). One study manipulated clause and case features as in (3-d). Lastly, 15 experiments from 12 studies manipulated only the clause feature as in (3-e).

- (3)
- The new executive_i [who oversaw the middle manager_j] apparently doubted *himself*_{i/*j} on most major decisions.
 - The tough soldier_i [that Fred_j treated in the military hospital] introduced *himself*_{i/*j} to all the nurses.
 - The son_i of the fireman_j hurt *himself*_{i/*j} in a bad accident.
 - The landlord_i told the janitor_j [that the fireman_k with the gas-mask would protect *himself*_{*i/*j/k} if it became necessary].
 - John_i thought [that Bill_j owed *himself*_{*i/j} another opportunity to solve the problem].

Importantly, the differentiation of the target and distractor on multiple structural/relational grounds give rise to alternative explanations that do not necessarily refer to the use of hierarchical relations between NPs as detailed as c-command in memory retrieval. One is that if the distractor appears in a different clause than the target, retrieval may proceed by finding the target antecedent that occurs in the same clause as the anaphor (Dillon et al., 2014; Wagers, 2008), which could well be driven by item-level features indexing temporal/contextual information (Balachandran, 2024).

Another alternative involves subjecthood. In all the studies that we reviewed the target antecedent was the subject of the clause containing the anaphor, and the only co-argument that preceded the anaphor. This uniform structural role raises the possibility that comprehenders of previous studies relied on a composite cue such as *subject-of-the-current-clause* in accessing the target (e.g., Kush and Phillips, 2014; Xiang et al., 2009). This cue may initiate a search for an antecedent that have the appropriate clausal index and subject features such as structural position or case marking (Arnett and Wagers, 2017; Avetisyan et al., 2020), without a need for full access to hierarchical organization of NPs. In either case, retrieval could succeed without appealing to fine-grained hierarchical relations among elements within a clause.

1.3. The current study

While numerous researchers have acknowledged the importance of testing the utility of relational notions like c-command in real-time processing, our review shows that it has proven difficult to uniquely isolate the role of c-command in the processing of anaphors. Our study here aims to fill this gap by directly testing whether hierarchical relations between NPs as in c-command guide antecedent retrieval. We use the preverbal, local reciprocal *birbirleri* in Turkish, which must be bound by a c-commanding, local (e.g., clause-mate) antecedent² (Akkus, 2021; Kornfilt, 1997). As we detail below, Turkish provides an ideal empirical ground to test this: (i) its rich morphological system allows us to manipulate the hierarchical availability of antecedents while eliminating the confounding cues that are commonly present in languages like English (i.e., clause, case and subjecthood), and (ii) its head-final structure permits the investigation of a preverbal anaphor without introducing a reactivation advantage for the target subject prior to the anaphor, thanks to the

²*Birbirleri* can also be bound by a non-c-commanding or non-local antecedent (see the examples in Kornfilt (1997); Paparounas and Akkus (2024)). However, these are often limited to certain syntactic or semantic environments, and are not relevant to the local binding conditions that we examine in this study.

post-anaphor establishment of the thematic relation between the verb and its subject (for a discussion of the importance of this point, see (Dillon et al., 2013; King et al., 2012)).

To anticipate the results, we found that c-commanding subjects *and* c-commanding objects were differentiated from non-c-commanding, clause-mate, similarly-case-marked distractors immediately at the point of processing the anaphor, regardless of their linear position in the sentence. This provides support for the view that hierarchical relations among NPs in a sentence, as captured in the configurational notion of c-command, are crucial for determining the availability of antecedents in early stages of memory retrieval above and beyond clause, case and subject features, and linear order.

2. Data Availability

The data, analysis scripts, experimental stimuli, and example recordings for the present study are available from <https://osf.io/7g3z8/files/osfstorage>.

3. Experiment 1: C-command and recency

Experiment 1 investigated the processing of the Turkish reciprocal *birbirleri*. We designed stimuli to isolate the role of c-command from other potential cues such as clause-mateness, case marking, topicality and linear order/recency. We controlled clause-mateness by having both the c-commanding target and the non-c-commanding distractor in the same clause; We controlled case-marking by using structures where both target and distractor were marked with the genitive case; We controlled topicality using a discourse context where neither target nor distractor were discourse prominent topics; And we factorially manipulated linear order of target and distractor to evaluate what impact it has on antecedent retrieval. We also factorially manipulated number match between the anaphor and the distractor to reveal the role of featural interference in antecedent retrieval (e.g., Badecker and Straub, 2002).

We used the visual world paradigm for spoken language comprehension in Experiments 1-3. This choice contrasts with many studies on processing anaphors, which generally employ reading methodologies. But while reading measures only generate indirect inferences about memory retrieval processes through variations in reading times (e.g., slow-downs or speed-ups), the visual world paradigm offers a more direct window into the competition between the target antecedents and distractors by tracking overt looks to their visual referents. It is also highly suitable for investigations of the real-time resolution of anaphors thanks to its high temporal resolution (e.g., Runner et al., 2003). This is due to the well-established assumption that attention

to visual objects in a visual display is closely time-locked to auditory input (Cooper, 1974; Tanenhaus et al., 1995), with a probabilistic relationship between fixations and the attention to the referents of visual objects that are actively engaged in linguistic processing (Allopenna et al., 1998; Magnuson, 2019).

We predict for Experiment 1 that if the c-command relation is rapidly used to guide antecedent retrieval above and beyond various cues as above, then there would be more looks to c-commanding targets than distractors across all conditions immediately within the reciprocal region. We further predict that if a cue-based retrieval mechanism is used to retrieve antecedents, this effect may be further moderated by number feature match and/or recency: More recent, feature matching distractors should be more likely to be erroneously retrieved at the critical anaphor (e.g., Jäger et al., 2015; Parker and Phillips, 2017; Sturt, 2003).

3.1. Materials

The stimuli in Experiment 1 were as in (4). The critical anaphor was the direct object of an embedded verb, and the target antecedent was the subject of that embedded clause. The distractor was a subordinated noun in the same embedded clause, i.e., embedded either in the subject NP as in (4-a) or in an adjunct NP as in (4-b). In hierarchical terms, the phrase of the target subject is a sister to the node containing the anaphor; hence, it c-commands the anaphor. However, the phrase of the distractor is not a sister to the same node; hence, it does not c-command the anaphor. The stimuli had a 2×2 within-subject design, crossing RECENCY of the target subject with respect to the distractor, i.e., Recent Target as in (4-a) and Distant Target as in (4-b), and number MATCH between the plural anaphor and the plural or singular distractor, i.e., Distractor Match and Distractor Mismatch, respectively. In all conditions, the only available antecedent that can bind *birbirleri* is the local, c-commanding referent, e.g., *kameramanlar* ‘cameramen’.

(4) Example item from Experiment 1 (with anaphor in bold and target in italics)

a. Recent Target, Distractor Match / Mismatch

Ayşe_i [yönetmen(-ler)-in_j yeni *kameraman-ları-nın*_k sertçe
 Ayşe director-PL-GEN new cameraman-3PL.POSS-GEN severely
birbirleri-ni_{*i/*j/k} eleştir-dik-leri]-ni anlat-tı.
 each.other-ACC criticize-NMLZ-3PL.POSS-ACC say-PST
 ‘Ayşe_i said that the director(s)’s_j new cameramen_k severely criticized each
 other_{*i/*j/k}.’

b. Distant Target, Distractor Match / Mismatch

Ayşe_i [*kameraman-lar-ın_j* yönetmen(-ler)-in_k
 Ayşe cameraman-PL-GEN director-PL-GEN
 ızn-ı-yle sertçe **birbirleri-ni**_{*i/j/*k}
 permission-3SG.POSS-COM severely each.other-ACC
 eleştir-dik-leri]-ni anlat-tı.
 criticize-NMLZ-3PL.POSS-ACC say-PST
 ‘Ayşe_i said that the cameramen_j, with the director(s)’s_k permission, severely
 criticized each other_{*i/j/*k}.’

In all conditions, both the target and the distractor were clausemates to the anaphor and marked with the genitive case *-(n)In*. Genitive case had different syntactic functions on the target and the distractor: When it was on the target, it marked the subject of an embedded clause; when it was on the distractor, it marked the possessor noun in a genitive-possessive construction.

We factorially manipulated the linear order/recency of the target relative to the distractor by using two different structures, as shown in (4). In both structures, a manner or time adverb, e.g., *sertçe* ‘severely’, was added before the critical anaphor to allow some distance between the last mentioned NP and the critical anaphor. To keep the saliency of the target and distractor comparable, we introduced a third referent as the discourse topic, i.e., *Ayşe*, in the matrix subject position.

Twenty-four sets of the four experimental conditions above were created and distributed across four lists in a Latin Square design. Before the visual world study, an offline antecedent selection task verified that *birbirleri* must be bound by the clausemate, c-commanding target antecedent, and that non-c-commanding distractors are not available as antecedents in the experimental stimuli, with them being selected 98% and 14% of the time, respectively (see Appendix B for details).

3.2. Participants

Sixty-eight native speakers of Turkish participated in the study ($M_{age} = 22$). They were undergraduate or graduate students at a public university in Istanbul, Turkey, and were recruited through social media. They had normal or corrected-to-normal vision. Each of them received 200 Turkish liras (~\$10.5 at the time of the study) for compensation.

3.3. Procedure

Before the experiment, the participants were informed about the procedure. This was followed by a consent process and a brief demographic survey. Participants were

tested individually in a private testing booth in a laboratory located at a public university in Istanbul, Turkey. Eye-movements were recorded via a desktop eye tracker, SR Eye Link 1000 Plus (SR Research, Ontario, Canada), with a sampling rate of 1000 Hz. Although viewing was binocular, only data from the right eye was recorded. They were rested on a chin rest to minimize head movements. Each session started with a calibration process, and further calibrations were done throughout the experiment as needed. There were three practice trials to familiarize the participants with the task. All instructions were given in Turkish. Each session took about an hour, and the participants were offered three breaks throughout the experiment.

In each trial in the study, the participants first listened to a lead-in sentence, then a question and lastly a critical sentence through headphones. An example lead-in sentence and question for the example critical sentence in (4) is in (5) below. The lead-in sentence provided background for the corresponding experimental sentence, and the question cued the participant to click on the referent of the reciprocal that would appear in the following sentence. The lead-in sentence included the nominal form of the embedded verb in the experimental sentence, and the question had the embedded verb itself to allow participants to anticipate the verb in the experimental sentence. Questions were given *before* the experimental sentence to direct participants' eye movements towards the referent of the anaphor as quickly as possible. This goal-oriented design allowed us to infer referential decisions from eye-movements during the processing of critical anaphor (Trueswell, 2008).

- (5) a. Lead-in
Ayşe maalesef birçok eleştiri-ye şahit ol-du.
Ayşe unfortunately several criticism-DAT witness-PST
‘Ayşe witnessed several criticisms unfortunately.’
b. Question
Peki Ayşe kim-in eleştir-il-diğ-in-i anlat-tı?
So Ayşe who-GEN criticize-PASS-NMLZ-3SG.POSS-ACC explain-PST
‘So who did Ayşe say was criticized?’

The screen was blank during the presentation of the lead-in sentence and the question. At the end of the question, a preview of the visual display corresponding to the critical sentence was presented, schematized in Figure 2. The display consisted of four images: the target and the distractor from the experimental sentence, and a singular and a plural referent that were not mentioned in the respective trial. During the preview, a frame highlighted each picture for 1 second, and participants were told

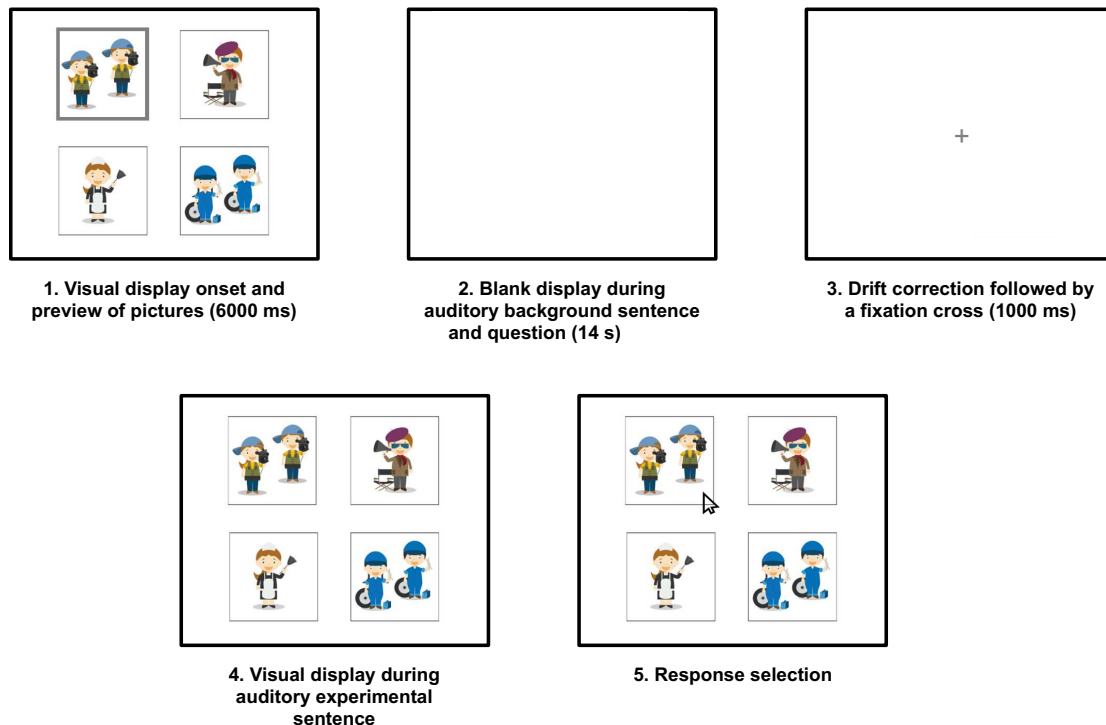


Figure 2: Schematization of experimental trials. In this example the visual display is for the stimuli in (4), with the corresponding target and distractor on the top left and right, and the unmentioned referents on the bottom.

to examine the images to familiarize themselves with the characters – this was done to limit eye-movements driven by visual features of the images to the preview session as much as possible (Henderson and Ferreira, 2004). The participants’ task was to click on the image of the referent that answered the preceding question. No feedback for accuracy was given. The participants’ eye-movements were only recorded during the presentation of the experimental sentences.

In addition to the experimental items, the study included 72 fillers that mirrored the structure of the experimental trials but replaced the anaphor with a name referent. This referent was either the distractor in the sentence (e.g., *yönetmen(ler)* ‘director(s)’ in (4)) in one-third of the fillers, the plural unmentioned referent in another third, or the singular unmentioned referent in the remaining third. This distribution ensured that each image had an equal likelihood of being mentioned just prior to the critical anaphor onset, preventing participants from predicting the

referent in advance.

The auditory stimuli were recorded by the first author, who is a native speaker of Turkish. Experimental sentences were first recorded in segments and then merged to ensure that there was a consistent onset for the critical anaphor across all conditions of a given item. Prior to the study, all visual stimuli were normed for codability (see Appendix D for further details).

3.4. Data analyses

The data were preprocessed using the VWPRE package in R (Porretta et al., 2022). All on-screen looks that did not fall on any of the four interest regions were recorded as outside. A dataset that was binned for time slices of 5 ms was created for plots and analyses. The binned dataset included proportions of looks to each interest region and outside for each time slice.

The main time window of interest for the analyses was the *reciprocal*, and its onset in each item set was treated as time zero in the analyses. An additional time window was the following word, the *embedded verb*. The duration of the two windows was the same across the conditions of an experimental item, but differed across items because of different case morphology on the reciprocal and lexical variation of the embedded verb. The average durations of the reciprocal and embedded verb in Experiment 1 were 649 and 782 ms, respectively. In the analyses, both time windows began at 150 ms post-stimulus onset to account for the delay in launching a saccadic eye movement in response to a stimulus (Rayner et al., 1983).

Two analyses for each critical time window were conducted. First, we analysed gaze probabilities to measure competition between the target vs. distractor. However, since this analysis could be influenced by precritical looking behavior, we also analyzed new fixations towards the target and the distractor that were launched within the critical time windows.

The data were analyzed in R, using mixed-effects linear or logistic regression models (Baayen et al., 2008). The *lmer* or *glmer* functions of the LME4 package were used (Bates et al., 2015). Fixed effects in Experiment 1 were RECENCY and MATCH, with random effects for participants and items. We fit two statistical models: a crossed and a nested model. The crossed model included both fixed effects and their interaction, and the nested model had a main effect of RECENCY and separate MATCH contrasts within the Recent Target and Distant Target conditions. Contrast coding for the fixed effects in both models are provided in Table 1. Each model included random intercepts by participants and by items, and random slopes when models converged (Barr et al., 2013). We report *t*- or *z*-scores, with an absolute value of 2 or above indicating significance at the α level .05 (Gelman and Hill, 2007).

	Crossed Model			Nested Model		
	Recency	Match	Recency:Match	Recency	Recent: Match	Distant: Match
Recent Target, Distractor Match	0.5	0.5	0.25	0.5	0.5	0
Recent Target, Distractor Mismatch	0.5	-0.5	-0.25	0.5	-0.5	0
Distant Target, Distractor Match	-0.5	0.5	-0.25	-0.5	0	0.5
Distant Target, Distractor Mismatch	-0.5	-0.5	0.25	-0.5	0	-0.5

Table 1: Contrast coding coefficients used for the crossed and nested statistical models in Experiment 1. Experimental conditions are given in rows, model contrasts given in columns.

3.4.1. Analysis of corrected log gaze probability ratios

Analyses of gaze probabilities in Experiment 1 were conducted on *corrected log target vs. distractor ratio*, computed separately for the reciprocal and embedded verb windows. This dependent variable compared the total gaze probability towards the target relative to the distractor that was corrected by the looks to the unmentioned referents. The formula for the variable is given in (6) (see Arai et al. (2007) for a similar measure). In (6) $P(\text{target}/\text{unmPL})$ refers to the proportion of looks to the target divided by looks to the unmentioned plural referent, and $P(\text{distractor}/\{\text{unmSG}, \text{unmPL}\})$ refers to looks to the distractor divided by looks to the unmentioned referent matching the distractor’s number (singular or plural). The final target vs. distractor ratio was calculated by subtracting the corrected looks to the distractor from those to the target on the log scale.

$$(6) \quad \ln(P(\text{target}/\text{unmPL})) - \ln(P(\text{distractor}/\{\text{unmSG}, \text{unmPL}\}))$$

Looks to the critical referents were normalized by looks to unmentioned referents to reduce the influence of visual competition in gaze patterns. Competitors with similar features can attract eye movements during visual search (Dahan and Tanenhaus, 2005), and in anaphor resolution, looks to the target or distractor may reflect both referential processing and such visual competition effects (Cunnings and Sturt, 2014). Since unmentioned referents are not linguistically relevant, we assume that looks to those would reflect visual competition only and use them as a baseline to isolate referential effects.

The ratios were calculated for each trial, for the reciprocal and embedded verb windows separately, and then entered into mixed-effects linear regression models.

3.4.2. New fixation analysis

For critical time windows shorter than 1 second such as those in the present study, the effect of an experimental factor that has begun in previous time windows such as $n-1$ or $n-2$ may persist in a following time window n , even if the factor no longer influences eye movements. Analyses on gaze probability ratios may thus be confounded with effects lingering from pre-critical windows, which may obscure

inferences on the timing of effects. To better estimate the time course of anaphora resolution, we also conducted analyses on the probabilities of new fixations that were initiated within the two critical time windows (see Kingston et al. (2016) for a similar analysis). We restricted this analysis to trials in which participants initiated a new fixation toward any of the four interest areas on the display. That is, trials with no fixation toward any of the four interest areas within the critical windows were excluded. We also did not count fixations toward an interest area if the relevant trial already started with looks to that area in the relevant window; so the analyses only included *new* fixations.

We conducted two analyses: one examining the probability of a new fixation on the target, and the other on the distractor, each relative to all new fixations. For each analysis, a binary dependent variable was created. In the former, trials with a new fixation on the target were coded as 1, and those without as 0. In the latter, trials with a new fixation on the distractor were coded as 1, and those without as 0. The two measures were calculated for each trial, for the reciprocal and embedded verb windows separately, and then entered into mixed-effects logistic regression models.

3.5. Results

An inspection of participants' choices in the filler items showed that they answered the questions correctly at least 70% of the time, indicating that they listened for comprehension. Trials that had a track loss of 50% or more were removed, which resulted in the loss of 3.2% of the data in experimental sentences. Data from two participants were removed from the analyses as they had 18 and 22 of their experimental trials excluded due to track loss. The remaining 66 participants' mean accuracy across filler items was 96.5%.

3.5.1. Picture selection data

Due to an error in coding the experiment, participants' choices in non-target answers were not recorded; so we only report whether or not they chose the target in Table 2. The results demonstrated that the target picture was overall chosen around 96.5% of the time.

A visual inspection of the data reveals a trend toward fewer target choices when it is linearly distant, with an additional decrease in the presence of a plural distractor. Since we cannot analyze the choices fully, we limit our interpretation to observable trends, and do not report inferential statistics.

3.5.2. Gaze probability analyses

Figure 3 plots the mean raw gaze probabilities towards the four interest areas in each condition, starting from the onset of the reciprocal and spanning a period of

	Target
Recent Target, Distractor Match	100%
Recent Target, Distractor Mismatch	99.7% (0.8)
Distant Target, Distractor Match	90.6% (1.7)
Distant Target, Distractor Mismatch	95.7% (1.3)

Table 2: Mean percentages of target picture selections (and Cousineau-Morey corrected, within-participant SEs following Cousineau (2005) and Morey (2008)) in Experiment 1.

1500 ms with time slices of 5 ms (see Appendix C for raw gaze probabilities for the entire sentence).

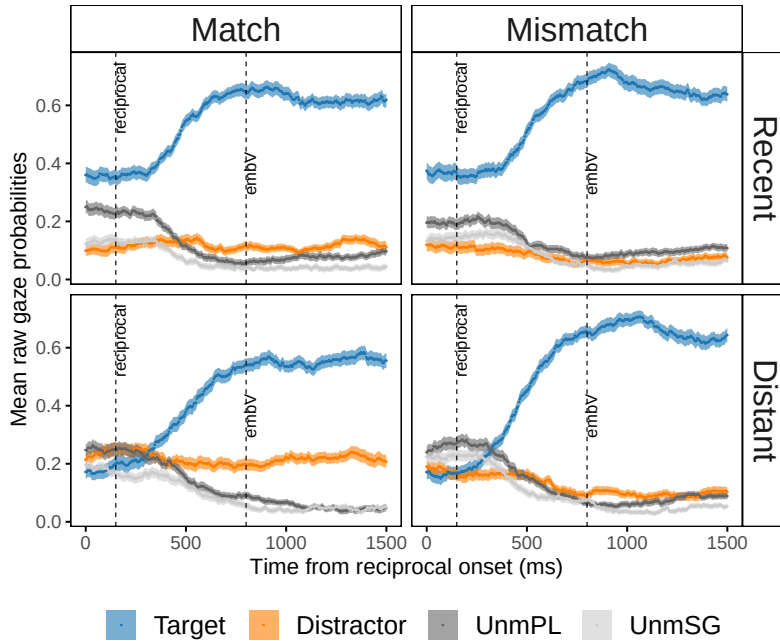


Figure 3: Mean raw gaze probabilities towards the four interest areas in each condition in Experiment 1. Error bands show Cousineau-Morey corrected, within-participant SEs. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Probabilities are time-locked to the onset of the critical anaphor and span 1500 ms. Dashed lines mark the 150-ms shifted onsets of the reciprocal and embedded verb (embV) windows.

Figure 4 shows the corrected log target vs. distractor ratios for the two critical time windows. As observed with positive target vs. distractor ratios in all conditions, there are overall more looks to the target than distractor. To reveal whether this is reliably modulated by any of the experimental factors, statistical analyses were

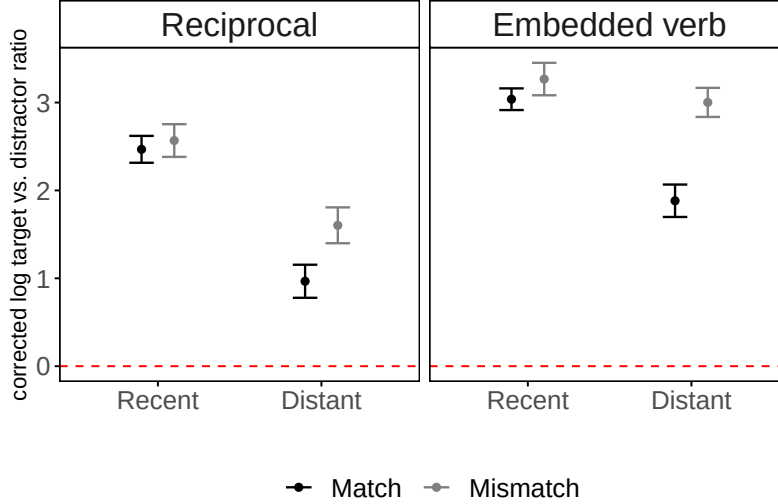


Figure 4: Mean of corrected target vs. distractor ratios on the log scale in each condition for the reciprocal and embedded verb windows in Experiment 1. Error bars show Cousineau-Morey corrected, within-participant SEs. Red dashed lines at 0 indicate equal looks to the target and distractor; values above 0 reflect greater looks to the target.

conducted, as reported below. The results of the statistical models on this measure³ are in Table 3.

In model output, the intercept is the estimated mean of corrected log target/distractor gaze probability ratio, averaged across the four experimental conditions. For the reciprocal window, the intercept estimate was well above 0 in both models, indicating overall more looks to the target relative to the distractor. The target received overall more looks when it was linearly more recent to the anaphor than the distractor. The nested model showed that in conditions in which the target was more distant, the distractor received more looks when it was plural compared to when it was singular. There was no such difference when the target was recent.

In the embedded verb window, there was again a clear, and even stronger, overall preference for the target. The target received overall more looks when it was more recent to the anaphor than the distractor. Moreover, plural distractors received overall more looks than singular ones, and there was also a significant interaction between RECENCY and MATCH of the distractor. The nested model gave insight into

³Model formulas for all the analyses on the visual world data in Experiments 1-3 are provided in Appendix E.

	Reciprocal			Embedded verb		
	Estimate	Std.error	t/z	Estimate	Std.error	t/z
<i>Corrected log target vs. distractor ratios</i>						
<i>Crossed model</i>						
(Intercept)	1.90	0.19	10.27	2.80	0.17	16.48
Recency	1.23	0.22	5.62	0.71	0.19	3.73
Match	-0.36	0.20	-1.80	-0.68	0.18	-3.69
Recency x Match	0.54	0.40	1.34	0.89	0.35	2.50
<i>Nested model</i>						
(Intercept)	1.90	0.19	10.25	2.80	0.17	16.48
Recency	1.23	0.22	5.62	0.71	0.19	3.71
Recent: Match	-0.09	0.25	-0.36	-0.24	0.23	-1.04
Distant: Match	-0.63	0.31	-2.05	-1.12	0.30	-3.76
<i>New fixations towards target</i>						
<i>Crossed model</i>						
(Intercept)	0.89	0.10	9.34	-0.24	0.07	-3.31
Recency	-0.01	0.16	-0.08	-0.18	0.16	-1.12
Match	-0.17	0.16	-1.09	-0.03	0.15	-0.22
Recency x Match	0.97	0.28	3.53	0.40	0.28	1.42
<i>Nested model</i>						
(Intercept)	0.90	0.10	9.28	-0.23	0.08	-2.77
Recency	0.03	0.16	0.21	-0.18	0.16	-1.13
Recent: Match	0.33	0.25	1.32	0.15	0.21	0.68
Distant: Match	-0.65	0.19	-3.47	-0.24	0.20	-1.20
<i>New fixations towards distractor</i>						
<i>Crossed model</i>						
(Intercept)	-1.27	0.10	-12.66	-0.88	0.11	-8.36
Recency	-0.26	0.14	-1.80	-0.39	0.16	-2.48
Match	0.35	0.18	1.98	0.37	0.20	1.83
Recency x Match	0.25	0.33	0.78	-0.14	0.37	-0.39
<i>Nested model</i>						
(Intercept)	-1.24	0.10	-12.88	-0.88	0.11	-8.19
Recency	-0.25	0.14	-1.76	-0.37	0.16	-2.39
Recent: Match	0.46	0.21	2.14	0.31	0.26	1.22
Distant: Match	0.20	0.19	1.06	0.47	0.27	1.75

Table 3: Coefficient estimates, and corresponding standard errors and t/z values for the fixed effects in mixed-effects linear/logistic regression models on corrected log target vs. distractor ratios and new fixations towards the target and distractor in Experiment 1. $|T/z|$ -values that are bigger than 2 indicate statistical significance, and are given in bold.

the nature of this interaction: As in the reciprocal window, when the target was more distant than the distractor, the distractor received more looks when it was plural compared to when it was singular; but there was no such difference when the target was more recent.

3.5.3. New fixation analyses

Figure 5 shows the proportions of new fixations towards the target and distractor out of all new fixations in each condition within the reciprocal and embedded verb windows. The results of the statistical models on the proportions are in Table 3.

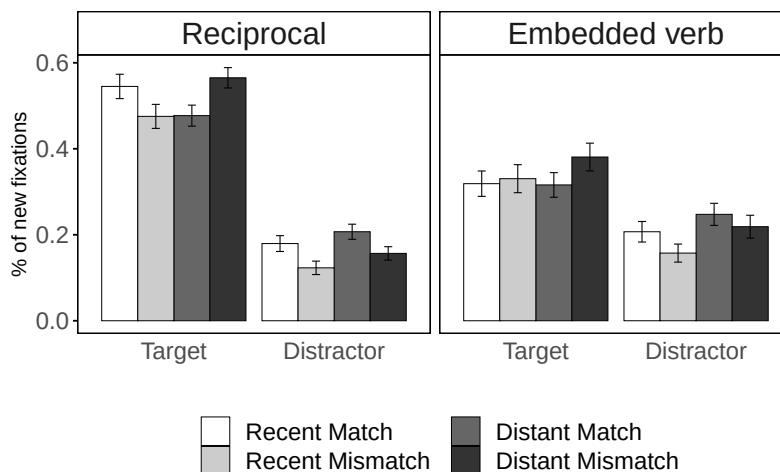


Figure 5: Mean percentages of new fixations towards the target and distractor in each condition within the reciprocal and embedded verb windows in Experiment 1. Error bars show Cousineau-Morey corrected, within-participant SEs.

In model output, the intercept estimate is the log odds of a new fixation towards the target or the distractor, averaged across the four experimental conditions. However, unlike in the gaze probability analyses, the intercept in these models does not test any hypothesis of interest, so although we report the intercepts for completeness, we do not directly interpret them. For new fixations towards the target, the analyses for the reciprocal window showed a significant interaction between RECENCY and MATCH. The nested model revealed the nature of this interaction: In conditions with a linearly distant target, there were fewer new fixations towards the target in the presence of a plural as opposed to a singular distractor, but the distractor number did not affect the fixations when the target was linearly recent. In the embedded verb window, the probability of a new fixation was not modulated by any of the factors.

For new fixations towards the distractor, the analyses for the reciprocal window showed a trend for more fixations towards a plural distractor, but this effect was not significant. The nested model revealed that this trend was driven by the recency of the target: In conditions with a linearly recent target, there were more fixations towards a plural distractor than a singular one, but there was no such difference when the target was distant. In the embedded verb window, there were fewer fixations towards the distractor when it was linearly distant (i.e., when the target was linearly more recent). There were no other effects.

3.6. Discussion

Experiment 1 investigated the role of the c-command relation in processing the local anaphor *birbirleri* in Turkish by employing stimuli that included c-commanding target subjects and non-c-commanding distractors in the same clause, marked with the same case marking. Despite neutralizing these potential cues to the target, the results showed that c-commanding subjects were immediately distinguished from distractors immediately at the reciprocal anaphor, as revealed by increased gaze probabilities towards the target over the distractor. This effect appeared regardless of whether the target linearly preceded or followed the distractor. Together, these findings suggest that antecedent retrieval is guided by fine-grained hierarchical information about NP relations within a sentence, as captured in the c-command relationship.

We also contrasted singular and plural distractors to reveal whether morphological number match between the anaphor and the distractor would modulate the retrieval process (e.g., Badecker and Straub, 2002). We found only limited evidence that number-matching distractors were reactivated during antecedent retrieval, as evidenced by the overall increased availability of target subjects over distractors. Moreover, the evidence for feature-based interference was not consistent, with effects varying depending on, but not consistently shaped by, the linear order of target and distractor. Specifically, we found a decrease in total gaze probabilities and probabilities of a new fixation towards target subjects in the presence of a plural, linearly more recent distractor. Conversely, we observed an increase in the probabilities of a new fixation towards the plural distractor when it was linearly more distant than the target. Given the mixed nature of these results, we sought to establish whether they would be replicated in Experiment 2 before interpreting them further.

Although we have highlighted the central importance of c-command in distinguishing the target from the distractor in our stimuli, this design alone does not unambiguously support the conclusion that c-command relation is the relevant relationship that guides early antecedent retrieval processes. One important caveat

is that in Experiment 1’s stimuli, the target antecedent was always the subject of the clause that included the anaphor. Thus, an alternative explanation could be that the target is retrieved through a combination of two item-level features, subject (e.g., genitive case) and clause features (e.g., local), without a need for exploiting hierarchical relations like c-command (although as Franck and Wagers (2020) note, ‘subject of the current clause’ is arguably a relational notion). This suggests that further work is needed to unambiguously establish c-command as a key factor in antecedent retrieval. We test this by examining the availability of c-commanding antecedents in object position in Experiment 2.

4. Experiment 2: C-command and subjecthood

Experiment 2 investigated whether the immediate retrieval of c-commanding antecedents in subject position observed in Experiment 1 extends to c-commanding antecedents in other syntactic positions; specifically those in indirect object position. In Experiment 1, and indeed, in all previous studies that we reviewed in the Introduction, the target antecedent appeared as the subject of the clause containing the anaphor. Thus it is possible that a composite cue like *subject-of-the-current-clause* underwrites antecedent retrieval for anaphors (see also Franck and Wagers (2020)). Under this scenario, the robust availability of target subjects immediately within the reciprocal region in Experiment 1 could reflect the use of subject and clause information rather than sensitivity to hierarchical relations between NPs.

However, local anaphors can be bound by arguments other than subjects in many languages including Turkish (Göksel and Kerslake, 2005; Kornfilt, 1997). A stronger test of the claim that c-command is used online to retrieve antecedents for anaphors could leverage this fact to test whether comprehenders also distinguish non-subject targets from distractors on the basis of c-command, rather than by grammatical role/subjecthood. To this end, in Experiment 2 we tested the availability of c-commanding targets in indirect object position, thereby disentangling the effect of argument role in the local clause from c-command. Like in Experiment 1, c-commanding targets and distractors were positioned in the same clause to eliminate clausal information as a distinguishing cue. Additionally, both the indirect object and the distractor were marked with dative case to neutralize case information.

We predicted that if the c-command relation between NPs, rather than subjecthood combined with clause information, guides antecedent retrieval, then we would observe increased looks to c-commanding subjects as well as c-commanding indirect objects when compared to non-c-commanding distractors.

4.1. Materials

Experiment 2 again employed Turkish sentences with *birbirleri* as the direct object of an embedded clause, as in (7). All sentences had a plural, c-commanding, clause-mate antecedent in the subject position and an additional dative-marked noun in the same clause. The stimuli had five conditions in a within-subject design. The conditions in (7-a)-(7-b) had a 2x2 design, crossing the C-COMMAND relation of the dative-marked noun with respect to the anaphor, i.e., c-commanding indirect object (IO) vs. non-c-commanding distractor, and number MATCH of this referent with the anaphor, i.e., singular vs. plural. In hierarchical terms, the phrases of the embedded subject and IO are a sister to the node containing the anaphor; hence, they c-command the anaphor. However, the distractor's phrase is embedded in a PP and is not a sister to the same node; hence, it does not c-command the anaphor. The fifth condition had a singular subject and a plural IO, as shown in (7-c), and was added as a control condition to investigate the overall availability of IO in the absence of a plural subject. In (7) we mark all c-commanding subjects and IOs as available antecedents through coindexation as they can occasionally bind the reciprocal regardless of their number (mis)match (see below).

(7) Example item from Experiment 2 (with anaphor in bold and target(s) in italics)

a. Plural Subject, IO Match / Mismatch

Ayşe_i [*kameraman-lar-ın_j yönetmen(-ler)-e_k açıkça **birbirleri-ni**_{*i/j/k}*
 Ayşe cameraman-PL-GEN director-PL-DAT openly each.other-ACC
 sor-duğ-u]-nu anla-dı.
 ask-NMLZ-3SG.POSS-ACC notice-PST

‘Ayşe_i noticed that the cameramen_j openly asked the director(s)_k about each other_{*i/j/k}.’

b. Plural Subject, Distractor Match / Mismatch

Ayşe_i [*kameraman-lar-ın_j yönetmen(-ler)-e_k göre açıkça*
 Ayşe cameraman-PL-GEN director-PL-DAT according.to openly
birbirleri-ni_{*i/j/*k} sor-duğ-u]-nu anla-dı.
 each.other-ACC ask-NMLZ-3SG.POSS-ACC notice-PST

‘Ayşe_i noticed that the cameramen_j, according to the director(s)_k, openly asked about each other_{*i/j/*k}.’

c. Singular Subject, IO Match

Ayşe_i [*kameraman-ın_j yönetmen-ler-e_k açıkça **birbirleri-ni**_{*i/j/k}*
 Ayşe cameraman-GEN director-PL-DAT openly each.other-ACC

sor-duğ-u]-nu anla-dı.
ask-NMLZ-3SG.POSS-ACC notice-PST
‘Ayşe_i noticed that the cameraman_j openly asked the director(s)_k about
each other*_{i/j/k}.’

Both the IO and the distractor followed the embedded subject, were clause-mates to the anaphor, and marked with the dative case *-(y)A*. While the dative noun formed an adjunct with the postposition, e.g., *göre* ‘according to’, in the distractor conditions, it functioned as the goal argument of the embedded verb in the IO conditions. The interpretation of the anaphor in the IO condition was ambiguous because there were two available antecedents in c-commanding positions: the embedded subject and the IO. There was only one available antecedent in the Distractor conditions: the embedded subject. The embedded verb, e.g., *sor-* ‘to ask’, was optionally ditransitive, allowing it to have two object arguments in the IO conditions and only a DO argument in the distractor conditions. A manner or time adverb, e.g., *açıkça* ‘openly’, was added before the anaphor in all sentences.

Thirty sets of the five conditions in (7) were created as experimental stimuli and distributed across five lists in a Latin Square design. Before the visual world study, an offline norming task confirmed that the experimental stimuli employed verbs forming configurations in which the indirect object c-commanded the critical anaphor in direct object position (Öztürk, 2005). Moreover, two offline antecedent selection tasks were conducted to verify that *birbirleri* can be bound by the clause-mate, c-commanding antecedents in subject and IO positions, but not the non-c-commanding distractors, in this set of stimuli. We note three key observations about the results (see Appendix B for details on the offline tasks). First, we found that plural IOs were more frequently selected as antecedents than singular IOs; confirming that c-commanding, plural IOs are available as antecedents. However, the overall selection rate of IOs was smaller than that of subjects (i.e., 35% vs. 91% & 20% vs. 85%), suggesting a subject bias in interpreting the reciprocal. Second, c-commanding, singular IOs were occasionally available as antecedents in the Plural Subject, IO Mismatch condition; and c-commanding, singular subjects were chosen more often than c-commanding, plural subjects in the control condition. These findings suggest that c-command and subjecthood may play a more dominant role in antecedent selection than number alone. While this is an important finding that deserves further investigation, our study focuses on testing the role of the c-command constraint; we thus proceeded with the stimuli above, which confirm that c-command is a crucial constraint in antecedent selection for *birbirleri*.

Third, when participants were explicitly prompted to consider all possible antecedents for the anaphor from the sentence as in a multiple-choice format, the overall rate of distractor selection was found to be high (i.e., 27%). However, a forced-choice task that followed showed that distractor selections were negligible (i.e., 2.2%) when participants were limited to a single antecedent choice. One explanation for the unexpectedly high selection of distractors in the multiple-choice format could lie in the postposition used. In all materials, the distractor was the complement of the postposition *göre* ‘according to’, which may take wide scope over the entire sentence (e.g., Ernst, 2001) and, in doing so, enable binding in the absence of a c-command relation. Given that local binding is still preferred over binding via wide scope in the forced-choice task, we proceeded with this stimuli for Experiment 2, but address this further by using a different set of postpositions in Experiment 3 (see below).

4.2. Participants

Fifty-four native speakers of Turkish ($M_{age} = 22.7$) from the same pool participated in the study. All participants had normal or corrected-to-normal vision. They received 300 Turkish liras ($\sim \$10$ at the time of the study) for compensation.

4.3. Procedure

Thirty sets of experimental sentences were interspersed with 72 filler items that had a similar format to those in Experiment 1. The rest of the procedure was the same as in Experiment 1.

4.4. Data analyses

The data in Experiment 2 were preprocessed and analyzed as in Experiment 1. The critical comparisons in the analyses differed from those in Experiment 1 in the following way. We performed separate statistical analyses for the 2x2 design, i.e., conditions with a plural subject as in (7-a)-(7-b), and the control condition, i.e., the Singular Subject with IO Match condition as in (7-c). For the 2x2 design, fixed effects were C-COMMAND and MATCH of the dative-marked noun, with random effects for participants and items. Both fixed effects were sum-contrast coded and included in a crossed and nested model that were fit to the data. The crossed model included both fixed effects and their interaction, and the nested model had a main effect of C-COMMAND and separate MATCH contrasts within the IO and Distractor conditions. Table 4 provides contrast coding for the fixed effects in both models for the 2x2 design. The rest of the model specifications were the same as in Experiment 1. The control condition was analyzed with a *t*-test comparing the looks to the singular subject vs. the plural IO. The average durations of the reciprocal and embedded verb in Experiment 2 were 632 and 630 ms, respectively.

	Crossed Model			Nested Model		
	C-command	Match	C-command:Match	C-command	IO: Match	Distractor: Match
IO Match	0.5	0.5	0.25	0.5	0.5	0
IO Mismatch	0.5	-0.5	-0.25	0.5	-0.5	0
Distractor Match	-0.5	0.5	-0.25	-0.5	0	0.5
Distractor Mismatch	-0.5	-0.5	0.25	-0.5	0	-0.5

Table 4: Contrast coding coefficients used for the crossed and nested statistical models for the 2x2 design in Experiment 2. Experimental conditions are given in rows, model contrasts given in columns.

4.5. Results

An inspection of participants’ accuracy in the filler items showed that two participants’ accuracy was below the 70% threshold, so they were removed from analyses. The remaining 52 participants’ mean accuracy across filler items was 93.3%. Trials that had a track loss of 50% or more were removed, which resulted in the loss of 0.008% of experimental items.

4.5.1. Picture selection data

The participants’ selections of pictures in the experimental items are given in Table 5. The results showed that overall the embedded subject was chosen around 81.2% of the time, IO 25.5% of the time and the distractor around 5.9% of the time.

A mixed-effects logistic regression model fit to the selection data for the dative-marked noun in the 2x2 design revealed main effects of C-COMMAND, with more choices of IO than distractor (z ’s ≥ 3.57); there were no other effects (z ’s ≤ 1.66). A binomial test that compared the proportion of subject and IO responses in the control condition showed that subject and IO responses were not significantly different ($p = 0.91$, 95% CI: 0.45 - 0.56%).

	Subject	IO/ Distractor	Unmentioned	
			Plural	Singular
Plural Subject, IO Match	82.1% (2.1)	16.3% (2)	0.6% (0.4)	1% (0.6)
Plural Subject, IO Mismatch	88.3% (2.4)	11.3% (2.3)	0%	0.3% (0.4)
Plural Subject, Distractor Match	91.8% (1.6)	6.9% (1.4)	0.3% (0.4)	1% (0.6)
Plural Subject, Distractor Mismatch	93.8% (1.9)	4.8% (1.8)	0.4% (0.4)	1% (0.6)
Singular Subject, IO Match	49.7% (3.3)	48.7% (3.5)	0.6% (0.4)	1% (0.6)

Table 5: Mean percentages of picture selections (and Cousineau-Morey corrected SEs) in Experiment 2.

4.5.2. Gaze probability analyses

Figure 6 plots the mean raw gaze probabilities towards the four interest areas on the display in each condition, spanning a period of 1500 ms with time slices of 5 ms

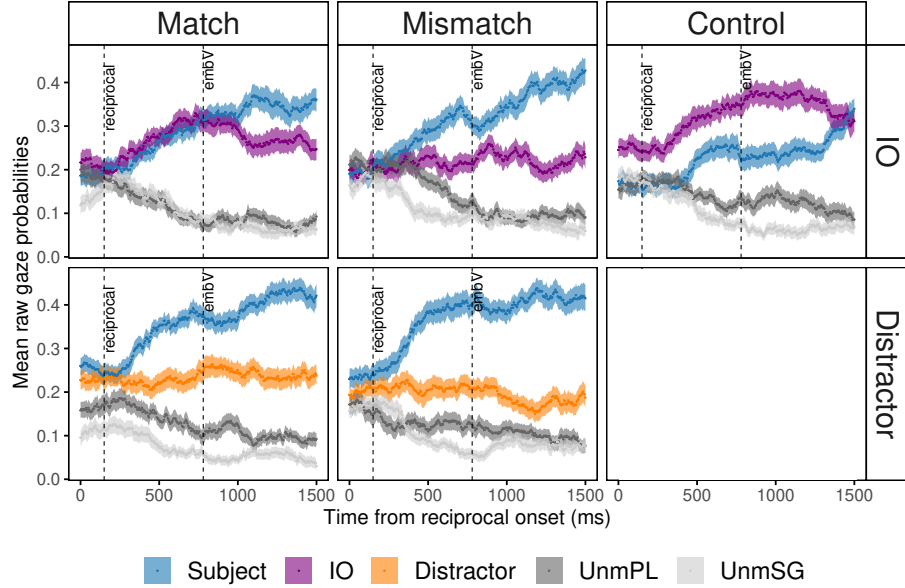


Figure 6: Mean raw gaze probabilities towards the four interest areas in each condition in Experiment 2. Error bands show within-participant SEs, with Cousineau–Morey correction normalized across all five conditions; the control condition was analyzed separately in the statistical models. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Probabilities are time-locked to the onset of the critical anaphor and span 1500 ms. Dashed lines mark the 150-ms shifted onsets of the reciprocal and embedded verb (embV) windows.

(see Appendix F for gaze probabilities for the entire sentence).

As in Experiment 1, the analyses were conducted on the corrected log subject vs. IO/distractor ratios that were calculated by subtracting the corrected gaze probability of IO or distractor from that of the subject on the log scale, for the reciprocal and embedded verb windows separately. Figure 7 shows this measure in each condition for the two time windows. Model output on this measure for the 2x2 design is in Table 6.

For the reciprocal window, there was a positive estimate in the intercept in both crossed and nested models, suggesting that there were overall more looks towards the subject than IO/distractor. This subject bias was around 2.5 times stronger in the embedded verb window than in the reciprocal. In both time windows, the subject bias was decreased when the dative noun was a c-commanding IO compared to when it was a distractor. There were no effects of number match or an interaction.

For the control condition, a *t*-test showed that the singular subject vs. plural IO ratio was not reliably different from 0, suggesting no bias towards the subject or IO

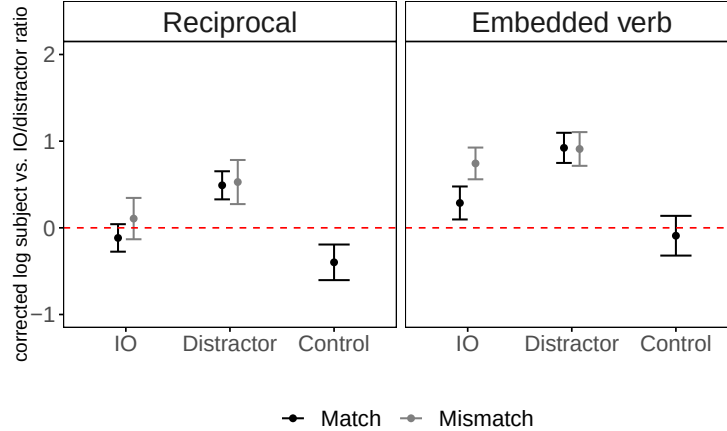


Figure 7: Mean of corrected subject vs. IO/distractor ratios on the log scale in each condition for the reciprocal and embedded verb windows in Experiment 2. Error bars show within-participant SEs, with Cousineau–Morey correction normalized across all five conditions; the control condition was analyzed separately in the statistical models. Red dashed lines at 0 indicate equal looks to the subject and IO/distractor; values above 0 reflect greater looks to the subject and those below 0 greater looks to the IO/distractor.

in the reciprocal ($t = -1.65$) or embedded verb window ($t = -0.39$).

4.5.3. New fixation analyses

Figure 8 shows the proportions of new fixations towards the subject and IO/distractor out of all new fixations in each condition within the reciprocal and embedded verb windows.

The analyses were run on the probabilities of new fixations towards the embedded subject and those towards the IO/distractor, for each time window. Model output on the proportions for the 2x2 design is in Table 6.

The analyses on new fixations towards the subject showed that the probability of a fixation was around 50% in both reciprocal and embedded verb windows, and this was not modulated by any of the experimental factors. For the analyses on new fixations towards the IO/distractor, the probability of a fixation was less than 50% in both time windows. In the reciprocal window, there were more fixations when the dative noun was plural, and there was a significant interaction between C-COMMAND and MATCH. The nested model showed that this interaction was driven by the c-commanding status of the dative noun: There were more fixations towards the plural than the singular, c-commanding IO, but there was no difference between singular and plural distractors. There were no other effects.

	Reciprocal			Embedded verb		
	Estimate	Std.error	t/z	Estimate	Std.error	t/z
<i>Corrected log subject vs. IO/distractor ratios</i>						
<i>Crossed model</i>						
(Intercept)	0.25	0.13	2.01	0.71	0.14	5.20
C-command	-0.51	0.23	-2.19	-0.40	0.20	-2.001
Match	-0.13	0.21	-0.63	-0.22	0.20	-1.09
C-command x Match	-0.17	0.43	-0.41	-0.48	0.40	-1.18
<i>Nested model</i>						
(Intercept)	0.25	0.13	2.01	0.71	0.14	5.16
C-command	-0.51	0.23	-2.19	-0.41	0.20	-2.02
IO: Match	-0.22	0.30	-0.73	-0.47	0.33	-1.40
Distractor: Match	-0.05	0.30	-0.16	-0.02	0.28	0.07
<i>New fixations towards subject</i>						
<i>Crossed model</i>						
(Intercept)	-0.08	0.08	-0.89	-0.16	0.08	-1.91
C-command	-0.16	0.15	-1.04	0.17	0.17	0.99
Match	-0.17	0.16	-1.09	-0.06	0.15	-0.40
C-command x Match	-0.27	0.34	-0.80	-0.45	0.34	-1.33
<i>Nested model</i>						
(Intercept)	-0.08	0.08	-0.93	-0.15	0.08	-1.84
C-command	-0.16	0.15	-1.04	0.18	0.18	1.02
IO: Match	-0.30	0.21	-1.44	-0.32	0.25	-1.27
Distractor: Match	-0.04	0.23	-0.18	0.16	0.21	0.74
<i>New fixations towards IO/distractor</i>						
<i>Crossed model</i>						
(Intercept)	-0.54	0.09	-6.18	-0.73	0.11	-6.80
C-command	0.29	0.17	1.68	0.11	0.17	0.64
Match	0.42	0.15	2.83	0.10	0.17	0.58
C-command x Match	0.77	0.32	2.41	-0.23	0.35	-0.66
<i>Nested model</i>						
(Intercept)	-0.54	0.09	-6.22	-0.73	0.11	-6.84
C-command	0.29	0.17	1.69	0.08	0.18	0.46
IO: Match	0.80	0.20	3.92	-0.04	0.29	-0.15
Distractor: Match	0.03	0.21	0.13	0.20	0.22	0.92

Table 6: Coefficient estimates, and corresponding standard errors and t/z values for the fixed effects in mixed-effects linear/logistic regression models on corrected log subject vs. IO/distractor ratios and new fixations towards subject, IO or distractor in Experiment 2. $|T/z|$ -values that are bigger than 2 indicate statistical significance, and are given in bold.

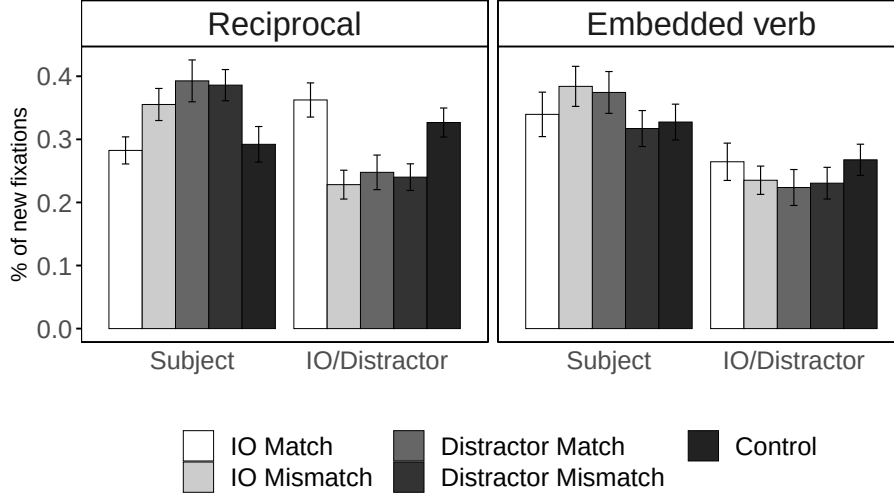


Figure 8: Mean percentages of new fixations towards the subject and IO/distractor in each condition within the reciprocal and embedded verb windows in Experiment 2. Error bars show within-participant SEs, with Cousineau–Morey correction normalized across all five conditions; the control condition was analyzed separately in the statistical models.

For the control condition, binomial tests compared the proportions of new fixations towards the subject vs. IO. Results showed that there were more fixations towards IO than the subject in the reciprocal window ($p < 0.01$, 95% CI: 0.52 - 0.67%); but fixations towards the subject and IO were not significantly different in the embedded verb window ($p = 0.64$, 95% CI: 0.44 - 0.60%).

4.6. Discussion

Experiment 2 tested whether the immediate availability of c-commanding subjects at retrieval extends to c-commanding indirect objects (IOs), a key prediction of the hypothesis that c-command is directly implicated in antecedent retrieval associated with anaphors. We observed that IOs were rapidly differentiated from non-c-commanding distractors at the reciprocal, despite having neutralized case and clause-hood cues. This provides evidence that retrieval targets antecedents that are in a hierarchically available position with respect to the anaphor, rather than approximating this relationship through the use of other structural cues such as *the-subject-of-the-current-clause* that do not necessarily refer to the hierarchical organization within the constituent structure of a sentence.

Interestingly, our results suggest that c-commanding singular antecedents, particularly subjects, can occasionally bind the reciprocal despite a number mismatch.

In offline antecedent selection tasks, in conditions comparing plural and singular IOs in the presence of plural subjects, plural IOs were selected more frequently than singular IOs, confirming the reciprocal’s sensitivity to number. This pattern was partly supported by the visual world data: While gaze probabilities did not differ between singular and plural IOs, participants made more new fixations to plural IOs in the reciprocal window. However, in the control condition comparing singular subjects and plural IOs, singular subjects were selected more often than plural IOs in antecedent selection tasks, with the visual world data showing no bias for either in gaze probabilities, but more new fixations toward the plural IO in the reciprocal window. Overall, these findings indicate a strong preference for c-commanding antecedents, especially for those in subject position, a preference that may potentially override number cues.

Before strongly interpreting this finding, however, we asked whether it would replicate in a higher-powered study (e.g. Nicenboim et al., 2018).

5. Experiment 3: High-powered, pre-registered replication

Experiment 3 aimed to determine whether the previously observed reactivation advantage for c-commanding subjects and IOs, compared to non-c-commanding distractors, would replicate in a higher-powered study.

To this end, we selected three conditions from Experiments 1-2 that compared the availability of c-commanding plural subjects followed by either (i) c-commanding plural IOs or (ii) non-c-commanding plural distractors. To maximize statistical power, only conditions involving plural targets and distractors were included. We compared genitive- or dative-marked distractors from Experiments 1 and 2, respectively, to examine whether the distractors’ differing case properties/structural role would impact antecedent retrieval beyond their syntactically subordinated status. Additionally, for the dative-marked distractors, we introduced new postpositions that do not permit the distractor to bind the reciprocal via a wide scope reading (see Experiment 2 above).

The experiment was pre-registered on OSF at the following link: <https://osf.io/s9f85>.

5.1. Materials

Similar to Experiments 1-2, Experiment 3 employed Turkish sentences in which *birbirleri* was the direct object of an embedded clause. All sentences had a c-commanding, clause-mate subject and an additional NP in the embedded clause, as exemplified in (8). The stimuli had three conditions, manipulating the C-COMMAND relation of this additional referent, i.e., c-commanding IO vs. non-c-commanding

distractor, and CASE marking on the distractor, i.e., dative vs. genitive. The conditions in (8-a)-(8-b) had the same structure as the Plural Subject, IO Match and Plural Subject, Distractor Match conditions from Experiment 2, respectively (see (7-a)-(7-b) above), and the one in (8-c) had the same structure as the Distant Target, Distractor Match condition from Experiment 1 (see (4-b) above). The conditions were relabeled here to have consistency within the experiment.

(8) Example item from Experiment 3 (with anaphor in bold and target(s) in italics)

a. IO

Ayşe_i [*yönetmen-ler-in_j* kameraman-lar-a_k açıkça **birbirleri-ni**_{*i/j/k}
 Ayşe director-PL-GEN cameraman-PL-DAT openly each.other-ACC
 sor-duk-ları]-nı anla-dı.
 ask-NMLZ-3PL.POSS-ACC understand-PST
 ‘Ayşe_i understood that the directors_j openly asked cameraman(s)_k about each other_{*i/j/k}.’

b. Distractor, Dative Case

Ayşe_i [*yönetmen-ler-in_j* kameraman-lar-a_k karşın açıkça
 Ayşe director-PL-GEN cameraman-PL-DAT despite openly
birbirleri-ni_{*i/j/*k} sor-duk-ları]-nı anla-dı.
 each.other-ACC ask-NMLZ-3PL.POSS-ACC understand-PST
 ‘Ayşe_i understood that the directors_j, despite the cameramen_k, openly asked about each other_{*i/j/*k}.’

c. Distractor, Genitive Case

Ayşe_i [*yönetmen-ler-in_j* kameraman-lar-ın_k yönlendirme-si-yle
 Ayşe director-PL-GEN cameraman-PL-GEN guidance-3SG.POSS-COM
 açıkça **birbirleri-ni**_{*i/j/*k} sor-duk-ları]-nı anla-dı.
 openly each.other-ACC ask-NMLZ-3PL.POSS-ACC understand-PST
 ‘Ayşe_i understood that the directors_j, with the cameramen’s_k guidance, openly asked each other_{*i/j/*k}.’

Both the IO and distractor followed the embedded subject and were clause-mates to the anaphor. IO was marked with the dative case $-(y)A$, and it was an argument of a ditransitive embedded verb, e.g., *sor-* ‘to ask’. The distractor was either marked with the dative case $-(y)A$ and followed by a postposition, e.g., *karşın* ‘despite’, forming an adverbial clause, or it was marked with the genitive case $-(n)In$ and

embedded in a genitive-possessive construction as a possessor noun. Whereas the c-commanding subject and the distractor had different case markings in (8-b), they were marked with the same genitive case in (8-c). For (8-b), two postpositions that lexically required dative case on their complement were used in the stimuli; *karşın* ‘despite’ and *rağmen* ‘despite’. Those postpositions are predicted to be less likely to take wide scope and bind the reciprocal, unlike the postposition *göre* ‘according to’ in Experiment 2 (see below for details). As in Experiment 2, the embedded verb, e.g., *sor-* ‘to ask’, was optionally ditransitive, allowing it to have two object arguments in the IO conditions and only a DO argument in the distractor conditions. The same verbs, which were normed for the hierarchical relationship between direct and indirect objects in Experiment 2, were used across conditions. A manner or time adverb preceded the anaphor, as in Experiments 1-2. The interpretation of the anaphor in the IO condition was ambiguous because there were two available antecedents in c-commanding positions: the embedded subject and the IO.

Twenty-four sets of the three conditions above were created and distributed across three lists in a Latin Square design. Before the visual world study, an offline antecedent selection task verified that *birbirleri* can be bound by the c-commanding subject or IO in this set of stimuli, with them being selected 95% and 46% of the time, respectively (see Appendix B for details). The overall low selection rate of distractors (i.e., 16%) verified that they were unavailable as antecedents to bind the reciprocal, also confirming their unavailability for binding via wide scope with the new set of postpositions (see Experiment 2 above).

5.2. Participants and power analysis

We ran a power analysis based on the data from Experiment 2 on the conditions in (8-a)-(8-b) to estimate the sample size of a high-powered study replicating the difference between plural IO and plural dative-marked distractor. We ran a model that examined new fixations towards the IO vs. distractor on a dataset that only included new fixations towards the subject and dative⁴. The results showed that there were more new fixations towards the IO than the distractor ($Est. = .78$, $SE = .34$, $z = 2.30$). We conducted 1000 simulations based on this model using the SIMR package in R (Green and MacLeod, 2016). Simulations showed that if the sample

⁴We also ran the same model on a dataset that included all new fixations, i.e., those towards all four regions, as we did in the main analyses. The results of this model also showed more fixations towards the IO than the distractor ($z = 2.66$), but this model had a bigger $|z|$ value than the one we report above. In order to be more conservative, we ran the power analysis based on the dataset that yielded a smaller z value.

size were increased to 102 subjects, a smaller size of the same effect ($\beta = .70$) would be replicated, with a power of 82.4% (95% CI: 79.9 - 84.7%).

We thus collected data from 102 native speakers of Turkish ($M_{age} = 22.1$) from the same pool for Experiment 3. None of the participants in the visual world study participated in the antecedent selection task above. All participants had normal or corrected-to-normal vision. Each participant received 300 Turkish liras ($\sim \$10$ at the time of the study) for compensation.

5.3. Procedure

In addition to the 24 sets of experimental sentences, there were 72 filler items that had the same format as those in Experiment 1. The rest of the procedure was the same as in Experiments 1-2.

5.4. Data analyses

The data in Experiment 3 were pre-processed and analyzed for total gaze probabilities and probabilities of a new fixation following the same steps in Experiments 1-2. Due to the differences in experimental conditions, the critical comparisons in the analyses differed from those in previous experiments in the following way. There was one fixed effect, second noun (NP2). This fixed effect was coded using Helmert contrasts, which compared the effects of the C-COMMAND relation of the second noun, i.e., IO vs. the two Distractor conditions, and CASE marking on the distractor, i.e., dative vs. genitive. Contrast coding for this fixed effect is given in Table 7. We fit a model with the fixed effect and random intercepts for participants and items, and random slopes when models converged. Analyses on gaze probabilities and new fixations were conducted as in Experiments 1-2. The average durations of the reciprocal and embedded verb in Experiment 3 were 659 and 752 ms, respectively.

	C-command	Case
IO, Dative	0.66	0
Distractor, Dative	-0.33	0.5
Distractor, Genitive	-0.33	-0.5

Table 7: Helmert contrast coding coefficients used for the mixed effects models in Experiment 3. Experimental conditions are given in rows, model contrasts given in columns.

5.5. Results

An inspection of participants' accuracy in the filler items showed that one participant's accuracy was below 70%, so they were removed from analyses. Overall

accuracy of the remaining participants was 95.5%. During preprocessing, trials that had a track loss of 50% or more were removed, resulting in the loss of 0.009% of the data for experimental sentences. Data from one participant were excluded due to track loss above 70% in the experimental sentences, leaving 100 participants for analysis.

5.5.1. Picture selection data

The participants’ selections of pictures in the experimental items are given in Table 8. The results showed that overall the subject was chosen around 87.5% of the time, the IO 20% of the time and the distractor around 8% of the time.

	Subject	IO/ Distractor	Unmentioned	
			Plural	Singular
IO, Dative	79.2% (1.7)	20.2% (1.7)	0.5% (0.2)	0.1% (0.1)
Distractor, Dative	92.9% (1.2)	6.6% (1.2)	0.4% (0.2)	0.1% (0.1)
Distractor, Genitive	90.4% (1.3)	9.2% (1.3)	0.4% (0.2)	0%

Table 8: Mean percentages of picture selections (and Cousineau-Morey corrected SEs) in Experiment 3.

A mixed-effects logistic regression model fit to the data for the NP2 choices revealed a main effect of C-COMMAND, with more choices of IO than distractor ($z = 4.51$). A comparison within the two distractor conditions showed that the genitive-marked distractor was chosen more often than the dative-marked distractor ($z = 2.06$). The increased choices of the genitive-marked distractors may reflect that participants occasionally considered these distractors as potential antecedents due to their case similarity to c-commanding subjects (see also Bakay (2020); Lago et al. (2019); Türk and Logačev (2024)). Regardless of this effect, the overall choices of the distractors were quite low.

5.5.2. Gaze probability analyses

Figure 9 plots the mean raw gaze probabilities towards the four interest areas in each condition, spanning a period of 1500 ms with time slices of 5 ms (see Appendix G for gaze probabilities for the entire sentence).

The dependent measure was the corrected log subject vs. IO/distractor ratios which were calculated by subtracting the corrected gaze probability of IO or distractor from that of the subject on the log scale, in each condition for the reciprocal and embedded verb windows, as illustrated in Figure 10 below. Model output on this measure is in Table 9.

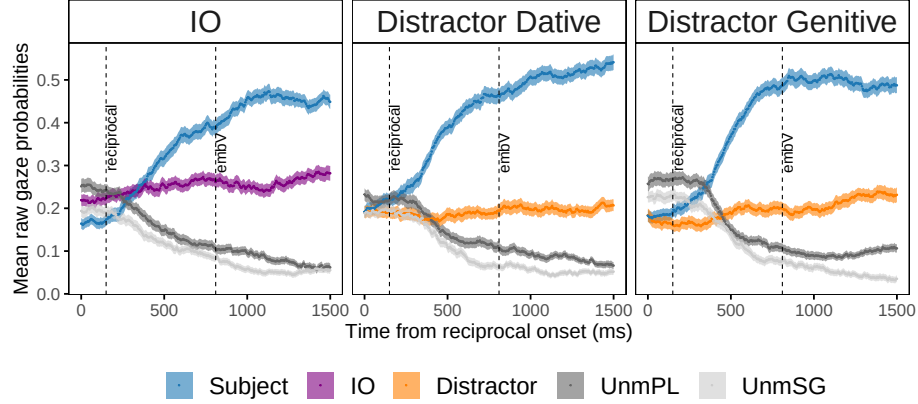


Figure 9: Mean raw gaze probabilities towards the four interest areas in each condition in Experiment 3. Error bands show Cousineau-Morey corrected, within-participant SEs. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Probabilities are time-locked to the onset of the critical anaphor and span 1500 ms. Dashed lines mark the 150-ms shifted onsets of the reciprocal and embedded verb (embV) windows.

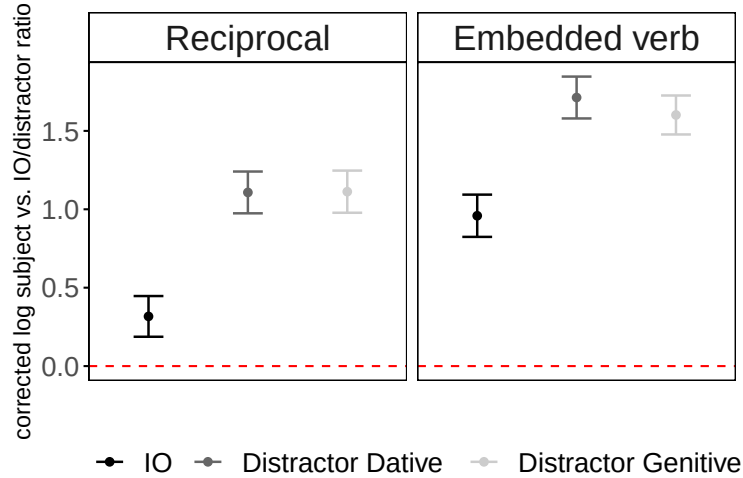


Figure 10: Mean of corrected subject vs. IO/distractor ratios on the log scale in each condition for the reciprocal and embedded verb windows in Experiment 3. Error bars show Cousineau-Morey corrected, within-participant SEs. Red dashed lines at 0 indicate equal looks to the subject and IO/distractor; values above 0 reflect greater looks to the subject.

In both reciprocal and embedded verb windows, there was a positive estimate in the intercept, suggesting overall more looks towards the subject over IO/distractor. The subject bias was stronger in the embedded verb window. In both time windows,

	Reciprocal			Embedded verb		
	Estimate	Std.error	<i>t/z</i>	Estimate	Std.error	<i>t/z</i>
<i>Corrected log subject vs. IO/distractor ratios</i>						
(Intercept)	0.84	0.12	6.91	1.42	0.14	10.34
C-command	-0.80	0.16	-5.08	-0.70	0.17	-4.17
Case	-0.003	0.19	-0.01	0.11	0.19	0.61
<i>New fixations towards subject</i>						
(Intercept)	0.24	0.11	2.19	-0.11	0.08	-1.31
C-command	-0.28	0.11	-2.65	0.11	0.11	0.93
Case	-0.03	0.12	-0.28	0.11	0.13	0.87
<i>New fixations towards IO/distractor</i>						
(Intercept)	-0.77	0.07	-11.60	-0.41	0.06	-7.04
C-command	0.33	0.13	2.60	0.05	0.13	0.42
Case	0.02	0.13	0.16	-0.05	0.13	-0.38

Table 9: Coefficient estimates, and corresponding standard errors and *t/z* values for the fixed effects in mixed-effects linear/logistic regression models on corrected log subject vs. IO/distractor ratios and new fixations towards the subject, IO or distractor in Experiment 3. $|T/z|$ -values that are bigger than 2 indicate statistical significance, and are given in bold.

this bias was decreased when the second NP was a c-commanding IO compared to when it was a distractor. There was no difference between the two distractor conditions.

5.5.3. New fixation analyses

Figure 11 shows the proportions of new fixations towards the subject and IO/distractor out of all new fixations in each condition within the reciprocal and embedded verb windows. Model output on the proportions is in Table 9.

For the analyses on new fixations towards the subject, the probability of a fixation was above 50% in the reciprocal window. There were fewer fixations towards the subject when the second noun was an IO compared to when it was a distractor. The overall probability of fixations towards the subject was around 50% in the embedded verb window and this was not modulated by either of the factors. For the analyses on new fixations towards the second noun, the probability of a fixation was less than 50% in both time windows. In the reciprocal window, there were more new fixations towards an IO than a distractor. There were no other effects.

5.6. Discussion

Experiment 3 was a high-powered, pre-registered replication of Experiments 1-2 that tested whether the immediate availability of c-commanding antecedents ob-

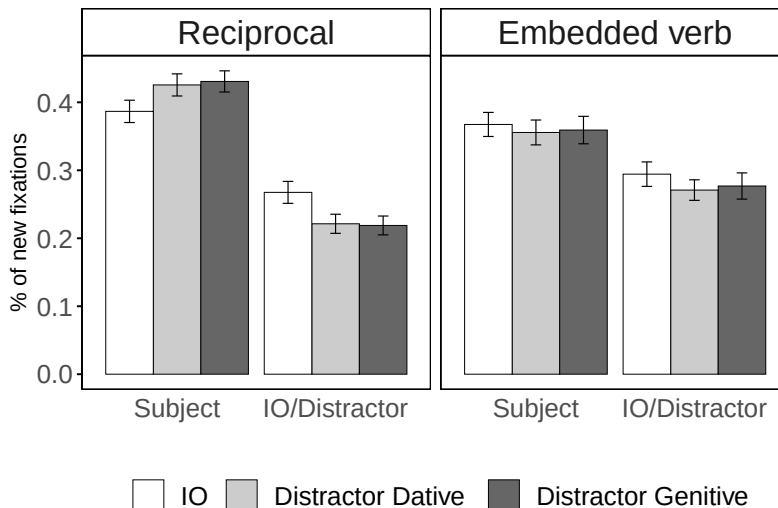


Figure 11: Mean percentages of new fixations towards the subject and IO/distractor in each condition within the reciprocal and embedded verb windows in Experiment 3. Error bars show Cousineau-Morey corrected, within-participant SEs.

served earlier would be replicated with a larger sample size. As in Experiments 1-2, the target antecedents and distractors were in the same clause, preventing the use of clause-level cues to distinguish them. Differently from Experiments 1-2, Experiment 3 directly compared two different distractors which differed in terms of their case similarity with the target and their structural role in the sentence.

The results of Experiment 3 replicated previous findings that c-commanding targets in subject and indirect object positions are differentiated from non-c-commanding distractors immediately within the reciprocal window by receiving more looks and attracting more new fixations. The decreased looks to the two distractors were comparable regardless of their case/structural roles. These findings suggest a robust and immediate role of hierarchical relations between NPs in antecedent retrieval above and beyond item-level structural cues such as case.

6. General Discussion

In this study we tested whether the initial stages of antecedent retrieval are mediated by hierarchical c-command relations by investigating the real-time resolution of the local, preverbal reciprocal anaphor *birbirleri* in Turkish. Across three visual world studies, we examined the time-course of reactivation of different potential antecedents for the critical anaphor. In Experiment 1, we saw overall more looks to

c-commanding subjects than case-matching, clause-mate distractors immediately at the reciprocal anaphor region. This effect was greater when the subject was linearly closer to the anaphor than the distractor. We also saw limited, inconsistent evidence for feature-based interference from plural distractors. In Experiment 2, we again saw overall increased looks to c-commanding subjects over non-c-commanding distractors, and further saw increased looks to c-commanding indirect objects relative to case-matching, clause-mate but non-c-commanding, distractors. Crucially, the analyses on new fixations revealed that more new fixations were launched towards the plural indirect object than the singular one, while the number feature of the distractor did not modulate new fixations. Experiment 3 was a pre-registered, high-powered replication of the key contrasts from Experiments 1-2; In Experiment 3 we again saw that c-commanding subjects and indirect objects were immediately reactivated at the anaphor to a larger degree than non-c-commanding distractors.

Overall, these findings provide novel evidence that hierarchical relations among NPs in a clause, as operationalized in the c-command constraint, is an important determinant of the availability of antecedents for local anaphors in early stages of memory retrieval. This extends previous observations by Cunnings et al. (2015); Franck and Wagers (2020); Kush et al. (2015); Moulton and Han (2018) with a novel methodology (visual world), and shows that the c-command advantage at retrieval extends to within clause hierarchical relations between NPs. Importantly, these results establish the importance of relational hierarchical structure above and beyond inherent syntactic features of an item in memory, such as clause, case and subjecthood, and domain-general memory factors such as linear order/recency, as we sought to design stimuli where these factors could not independently explain the current findings. Finally, these findings also extend previous observations by showing that non-subject NPs (here, indirect objects) that have the key property of c-command have a similar advantage at retrieval to c-commanding subjects.

We have adopted the notion of *c-command* (Reinhart, 1976) to characterize the hierarchical constraint on the relations between the anaphor and its antecedent for perspicuity. There are, however, alternative formalizations of relational constraints between NPs in a clause (e.g., Bruening, 2014; Chomsky, 1981; Langacker, 1969; Lasnik, 1976). While these formalizations may differ in important theoretical aspects, such differences do not directly bear on the interpretation of our findings and are not relevant to our discussion on their use in real-time sentence comprehension.

Overall, then, our results are consistent with a tight relationship between the hierarchical relational constraints on a given dependency (here, reciprocal anaphors), and the real-time reactivation of potential antecedents at the point of processing a dependent anaphor (Kush, 2013). It appears that something like the c-command

constraint is immediately available to guide antecedent retrieval processes, rather than this being a second-order, inferred relationship between items in memory, which would imply that it should become available relatively late in processing (McElree and Doshier, 1993). In what follows, we discuss the consequences of this observation for psycholinguistic theories of dependency formation.

6.1. Hierarchical information in memory retrieval

How could a hierarchical relationship like c-command be used in online interpretation in language comprehension? One possibility is to maintain the assumption that hierarchically available, c-commanding antecedents are differentiated from unavailable distractors at the point of retrieval, through a mechanism that leverages hierarchically informed and morphological (e.g., number) features as retrieval cues in reactivating the target, as in cue-based activation models (Lewis and Vasissth, 2005; Lewis et al., 2006). In the current setting, this means that upon hearing the reciprocal, all items in memory (e.g., c-commanding targets and non-c-commanding distractors) will be reactivated as a function of their match with retrieval cues and accessed in parallel; and c-commanding items will eventually receive a higher level of reactivation due to their match with hierarchical in addition to featural cues.

However, articulating this view in finer detail is challenging: It is not clear how to integrate hierarchical relational constraints as retrieval cues in a cue-based system (Alcocer and Phillips, 2012; Kush, 2013; Kush et al., 2015). As discussed above, a single item’s representation may not easily include relational information between items as in *x c-commands y* nor can they explicitly encode hierarchical positional information due to the exclusive use of feature-based information (e.g., Alcocer and Phillips, 2012; Dillon, 2011; Kush, 2013; Wagers, 2008).

To address this challenge, it has been proposed that item representations may be enriched with features that *approximate* hierarchical relations, rather than directly encode them (Dillon et al., 2014; Kush, 2013; Kush et al., 2015). One such implementation was proposed by Kush (2013) for representing the c-command relation within clauses: C-commanding items within a clause are predictively marked as availables based on the evolving phrase structure constructed by the parser. In particular, noun phrases that are either directly attached to the spine of the currently processed clause (i.e., branches immediately dominated by the clause node in the syntactic tree) or serve as its arguments are assigned the feature LOCAL:1, while other items receive LOCAL:0. This feature is dynamically updated during parsing such that, at the onset of a new clause, all LOCAL:1 values from the previous clause are rewritten as LOCAL:0. Although the LOCAL feature does not explicitly encode c-command in memory representations, it allows the parser to approximate the rel-

evant relational constraints of local anaphors by dynamically guiding retrieval (see Kush 2013 for extended discussion, and concerns about the generalizability of this feature). In essence, this model maintains a strictly cue-based position on memory access during real-time dependency formation, and reconciles this with the hierarchical nature of language comprehension by proposing that the feature content of items is dynamically managed so that all and only the hierarchically accessible items in memory at a given point have the relevant LOCAL feature.

Assuming hierarchically informed features like LOCAL, standard cue-based retrieval models predict that all cues, including hierarchical (e.g., LOCAL) and morphological ones (e.g., PLURAL), are combined in an equally weighted, linear fashion in reactivating the antecedents. Alternative accounts suggest that cues can be combined in a differentially weighted fashion, with hierarchical ones being prioritized over others; thereby maximizing the reactivation of the target while minimizing the one of the distractor (e.g., Kush and Phillips, 2014; Parker and Phillips, 2017) (see Dillon et al. (2013); Sturt (2003); Van Dyke and McElree (2011) for a stronger view). Both accounts predict partial reactivation of feature-matching distractors; with this effect predicted to be substantially smaller in the differentially weighted cue combination account than in the linear account. Given that we saw relatively little interference, our findings are more compatible with a retrieval model in which hierarchical cues are weighted more heavily than others, ensuring early and efficient access to the target antecedent of the anaphor. In such a model, distractors may still be partially reactivated due to feature match, but this may be too subtle to be reliably detected in medium-sized experiments (Jäger et al., 2015; Jäger et al., 2020), possibly accounting for the inconsistent pattern for interference in our findings.

6.2. *Hierarchical information in memory representations*

An alternative to reconciling hierarchical structure with a cue-based retrieval mechanism is to propose instead that hierarchical information shapes how linguistic items are organized and maintained in short-term or working memory. On this view, c-commanding items are rapidly accessed not because of retrieval cues targeting them, but because of how they are stored and represented within memory.

Recent structurally oriented accounts of sentence processing propose that linguistic items (e.g., words) are maintained through transient associations with their hierarchical positions, and that retrieval is guided by cues linked to these positions (Keshev et al., 2025). These item-to-position associations may also serve a broader organizational function, enabling memory systems to narrow the set of accessible items and improve efficiency for cognitive operations.

From this perspective, it could be that task-relevant, hierarchically prominent items such as locally c-commanding subjects and indirect objects may be stored in temporarily active, dynamically updated, privileged memory addresses that facilitate rapid access to them in retrieval. In sentence processing, all actively processed items within a sentence are considered to be reactivated from long-term memory (e.g., McElree, 2000). Building on this, we suggest that there may be an additional state of information in memory: a spatially structured memory state that temporarily binds content (e.g., lexical items) to context (e.g., structural template for the c-command domain), yielding a limited set of elements immediately available for direct access through their current context information, without requiring a search in memory or the postulation of a distinct ‘working memory’ store (e.g., Foraker and McElree 2007; McElree 2006) (see Oberauer (2002, 2009) for a similar component in his general-purpose working memory architecture, i.e., *the region of direct access*). When c-commanding items are the object of a cognitive operation, i.e., the antecedent of a dependent anaphor, they might thus be rapidly accessed thanks to their privileged association with this structured region of memory. This account differs from cue-based retrieval models as it attributes rapid access to targets not to their reactivation due to matching features, but to their temporary representational association with privileged structural roles, which results in direct access to them. Consequently, this view does not predict distractors to be reactivated or be accessible and interfere during early stages of retrieval, as they lack this representational advantage. However, the activation of feature-matching distractors may still accumulate in time, potentially leading to increased attention or interference in later stages of processing, as observed in some studies (e.g., Kush and Phillips, 2014; Sturt, 2003).

The primary goal of this study was to test the role of hierarchical relations in antecedent retrieval in memory, rather than to determine the exact mechanism underlying the implementation of hierarchical constraints. To this end, we conducted analyses on aggregate patterns within the anaphor window. The lack of consistent evidence for the reactivation of feature-matching distractors in this window may point towards a privileged representation for hierarchically available items. However, we caution against drawing this conclusion at the moment because it is also possible that we have not been able to observe reactivation as it was too brief or transient to be detected in the overall time window. Disentangling these two accounts would require a fine-grained analysis over the time course of access to critical referents or computational simulations to better evaluate the predictions of the two accounts (see Nicenboim and Vasishth (2018)). We leave this for future work.

It is important to note that neither of the accounts discussed above offers a mechanism for directly encoding an abstract, relational representation — specifically one

that captures the c-command relation between its two relata. Instead, the availability of c-commanding items is modeled by virtue of their hierarchically prominent position, implemented either through anticipatory feature assignment or through a reserved, structurally privileged store. The rapid use of relational information in memory retrieval may require relational representations, the precise nature of which remains to be clarified within psycholinguistic theories of dependency formation in future work (for relational representations in the domain of visual processing, see (Hafri and Firestone, 2021)).

One possible interpretation of our findings to bear in mind is that the immediate availability of c-commanding subjects and indirect objects may reflect a retrieval mechanism that draws on argument structure information (Wagers, 2008), which some theories identify as central to anaphor binding (Pollard and Sag, 1994; Reinhart and Reuland, 1993). In our stimuli, c-commanding antecedents always functioned as co-arguments of the anaphor. Argument structure may guide antecedent retrieval for preverbal anaphors such as *birbirleri*, as their co-arguments can be identified via case marking or the emerging syntactic parse of the sentences. In addition, c-commanding arguments in our stimuli had not yet been integrated into a dependency at the point of the anaphor since their head, i.e., the verb, is yet to be encountered. As such, their status as ‘open, yet-to-be-integrated’ arguments may have further increased their availability during antecedent retrieval. To test these hypotheses, future work should examine access to hierarchically available antecedents that are not syntactic co-arguments of the critical anaphor and/or use configurations in which anaphor appears in a non-argument position, removing the antecedents’ co-argument status.

6.3. Subject bias

While our findings indicate that comprehenders immediately reactivate both c-commanding subjects and indirect objects upon processing the Turkish reciprocal, there was a consistent bias for the subject, suggesting a role for argument role/thematic status in accessing the antecedent above and beyond c-command. This was evident with a higher rate of subject choices in the offline tasks and increased attention to the subject over the indirect object in the visual world studies. The increased availability of subjects may be due to their more prominent ranking as possible antecedents for anaphoric binding, as formulated in semantically oriented argument hierarchies (Jackendoff, 1972; Pollard and Sag, 1994). Additionally, the early increase in the visual attention to subjects may be, at least partly, due to the crosslinguistically observed agent preference that is rooted in the central role that agents play in sentence comprehension (Schlesewsky and Bornkessel, 2004) and more broadly in cognition (Leslie, 1995). That is, comprehenders often have a tendency

to identify and track agents as they guide the interpretation of the events being constructed (e.g., Cohn and Paczynski, 2013; Galazka and Nyström, 2016; Webb et al., 2010). Lastly, while we assessed the overall availability of indirect objects as binders, we did not control for the competition between subject and indirect object interpretations of the anaphor, which, in some cases, may have been biased toward the subject due to the semantic content of the sentences. Future studies that better control for this competition could reveal the extent of the subject bias that we observed.

7. Conclusion

In sum, we examined the role of item-to-item, hierarchical relations between noun phrases within a clause as in *x c-commands y* in antecedent retrieval using the local reciprocal in Turkish. In three visual world studies, we measured the moment-by-moment reactivation of c-commanding subject and indirect object antecedents, contrasted with non-c-commanding distractors. The stimuli were carefully constructed to distinguish the role of c-command relations from item-specific structural information (i.e., clause, case and subjecthood) and linear order/recency. Results consistently showed that c-commanding targets (i.e., subjects and indirect objects) were immediately distinguished from distractors upon encountering the reciprocal. These findings provide robust evidence that hierarchical relational information guides antecedent retrieval above and beyond other sources of structural information and linear order at the earliest stages of processing a local anaphor. The immediate availability of c-commanding items can be captured within cue-based activation models in which item representations are equipped with hierarchically informed features that are cued in retrieval (Kush, 2013). Alternatively, we suggest that this finding may be modeled with memory architectures that prioritize hierarchically prominent items for direct access during retrieval. Despite these two accounts, more work is required both to better evaluate their predictions and to determine whether, and how, models of real-time sentence comprehension can directly capture abstract relational representations.

Ethics approval

All experimental procedures were approved by the UMass Institutional Review Board (Protocol No.3977; approved on December 7, 2022).

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Appendix A. A review of studies on local anaphor processing

Citation	Language	N	Embedding of target or distractor	C-commanding of distractor?	Clause-mate distractor?	Case matching? distractor?	Argument role of distractor	Recency of target	Matching feature of distractor	Position of anaphor	Target match effects?	Target vs. distractor bias	Distractor match main effects?	grammatical sentences?	Distractor match effect in ungrammatical sentences?
<i>Event-related potentials</i> Kiang et al. (2009) (reflexives)	English	34	RC	No	No	Yes	Subject	Distant	Gender	Postverbal	Yes	-	-	-	No
<i>Eye-tracking-while-reading</i> Andrews (2021) E2 (reflexives)	English	64	PP	No	Yes	No	Possessor	Distant	Number	Postverbal	cri,-1 (L)	-	No	N/R	No
Cummings and Felser (2013) E1 (low WM)	English	16	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	cri (E, L); cri,+1 (L)	-	cri+2 (E)	No	No
Cummings and Felser (2013) E1 (high WM)	English	16	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	No	-	cri+2 (E)	No	No
Cummings and Felser (2013) E2 (low WM)	English	16	RC	No	No	Yes	Subject	Distant	Gender	Postverbal	cri+1 (L)	-	No	cri (FAC, E)	No
Cummings and Felser (2013) E2 (high WM)	English	16	RC	No	No	Yes	Subject	Distant	Gender	Postverbal	cri+1 (L)	-	No	No	No
Cummings and Sturt (2014) E1	English	28	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	cri,+1 (L)	-	No	N/R	N/R
Dillon et al. (2013) E1 (reflexives)	English	40	RC	No	No	No	Object	Distant	Number	Postverbal	cri (E,L)	-	No	No	No
Dillon et al. (2013) E2	English	32	RC	No	No	No	Object	Distant	Number	Postverbal	cri,+1 (L)	-	No	N/R	N/R
Felser et al. (2009) E2b (native, c-command)	English	37	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	-	-	-	No	-
Felser et al. (2009) E2b (native, no c-command)	English	37	SC	No	No	No	Nonsubject	Recent	Gender	Postverbal	-	-	-	No	-
Jäger et al. (2015)b E1	Chinese	150	AC	No	No	-	Subject	Distant	Animacy	Postverbal	cri (E,L); cri,-1,+1 (L)	-	cri-1 (L); cri,+1 (E, L)	No	cri-1 (INH, L); cri (INH, E, L); cri+1 (INH, E)
Jäger et al. (2015)a E2	German	151	RC	No	No	Yes	Subject	Distant	Gender	Preverbal	-	-	-	No	-
Jäger et al. (2015)a E3 (reflexives)	Swedish	35	RC	No	No	Yes	Subject	Distant	Gender	Postverbal	-	-	-	No	-
Jäger et al. (2020) (reflexives)	English	190	RC	No	No	No	Object	Distant	Number	Postverbal	N/R	-	N/R	cri (INH, E)	cri (FAC, L)
King et al. (2012) E2	English	48	RC	No	No	No	Object	Distant	Gender	Postverbal	cri (E, L); cri,+1 (E)	-	cri (E,L)	N/R	N/R
Parker and Phillips (2017) E1 (1-feature-mismatch)	English	30	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	cri,-1 (L)	-	No	N/R	No
Parker and Phillips (2017) E1 (2-feature-mismatch)	English	30	SC	Yes	No	Yes	Subject	Recent	Gender & Animacy	Postverbal	cri (L)	-	No	N/R	cri (FAC, L)
Parker and Phillips (2017) E2 (reflexives)	English	30	RC	No	No	Yes	Subject	Distant	Number & Animacy	Postverbal	-	-	No	N/R	cri (FAC, E, L); cri+1 (L)
Parker and Phillips (2017) E3 (1-feature-mismatch)	English	24	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	cri (L)	-	No	N/R	No
Parker and Phillips (2017) E3 (2-feature-mismatch)	English	24	SC	Yes	No	Yes	Subject	Recent	Gender & Number	Postverbal	cri,-1 (L)	-	No	N/R	cri (FAC, E, L)
Patil et al. (2016)	English	40	RC	No	No	Yes	Subject	Distant	Gender	Postverbal	cri (L)	-	N/R	cri (INH, E)	No
Sezer (2020) E3	Turkish	40	RC	No	No	No	Subject	Recent	Animacy	Preverbal	cri+1: (E); cri,+1,+2,+3 (L)	-	cri,+1,+2 (L)	cri,+2 (INH, L)	cri+1 (INH, L)
Seszer (2020) E4	Turkish	30	AC	No	No	Yes	Subject	Distant	Animacy	Preverbal	cri,+1,+2 (L); cri+3 (E)	-	cri (E, L)	cri (INH, L)	cri (INH, E)
Slogett (2017) E1b (speech verbs)	English	37	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	cri,+1 (L)	-	No	cri (FAC, E, L)	cri (FAC, E, L)
Slogett (2017) E1b (perception verbs)	English	37	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	cri,+1 (L)	-	No	No	cri+1 (FAC, E)
Slogett (2017) E2	English	45	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	cri,+1 (L)	-	cri,+1 (L)	No	cri+1 (FAC, L)
Sturt (2003) E1	English	24	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	cri (E,L); cri,+1,+2 (L)	-	cri+2 (L)	cri+2 (FAC, L)	No
Sturt (2003) E2	English	24	RC	No	No	No	Object	Distant	Gender	Postverbal	cri (E,L); cri+1 (L)	-	No	N/R	N/R
<i>Eye-tracking-while-listening</i> Clackson et al. (2011) E2 (control, reflexives)	English	40	AC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	-	Yes	-	No	-
Clackson and Heyer (2014)	English	42	AC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	-	Yes	-	Yes	-
Han et al. (2021) E2 (reflexives)	English	28	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	-	Yes	-	No	-
Runner and Head (2014) (reflexives, SRC)	English	25	RC	No	No	No	Object	Distant	Gender	Postverbal	-	Yes	Yes	Yes	-
Runner and Head (2014) (reflexives, ORC)	English	25	RC	No	No	Yes	Subject	Distant	Gender	Postverbal	-	Yes	Yes	No	-
<i>Priming</i> Nicol (1988) E1 (reflexives)	English	61	SC	Yes	No	Yes/No	Subject/Object	Recent	Gender	Postverbal	-	Yes	-	No	-
<i>Self-paced reading</i> Badecker and Straub (2002) E3 (reflexives)	English	32	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	-	-	-	cri+2 (INH)	-
Badecker and Straub (2002) E4	English (recip)	36	SC	Yes	No	Yes	Subject	Recent	Number	Postverbal	-	-	-	cri+1-4 combined (INH)	-
Badecker and Straub (2002) E5 (reflexives)	English	28	NP	No	Yes	No	Possessor	Recent	Gender	Postverbal	-	-	-	No	-
Badecker and Straub (2002) E6	English	48	SC	No	No	No	Nonsubject	Recent	Gender	Postverbal	-	-	-	No	-
Chen et al. (2012)	Chinese	120	AC	No	No	-	Subject	Distant	Animacy	Postverbal	-	-	-	cri,+1 (INH)	-
Clifton et al. (1999) E1 (reflexives)	English	48	PP	No	Yes	No	Possessor/Adjunct	Distant	Number	Postverbal	-	-	-	No	-
Clifton et al. (1999) E2 (reflexives)	English	48	PP	No	No	No	Possessor/Adjunct	Distant	Number & Gender	Postverbal	-	-	-	No	-
Clifton et al. (1999) E3 (reflexives)	English	49	SC	Yes	No	Yes	Subject	Recent	Gender	Postverbal	-	-	-	No	-
Jäger et al. (2015)a E1	German	144	RC	No	No	Yes	Subject	Distant	Gender	Preverbal	-	-	-	No	-
King et al. (2012) E1	English	48	RC	No	No	No	Object	Distant	Number	Postverbal	cri+1	-	No	N/R	N/R
Kush and Phillips (2014)	Hindi (recip)	32	RC	No	No	No	Object	Distant	Number	Preverbal	cri+1	-	No	No	No
Laurinavichyute et al. (2017) E1 (reflexives)	German	111	RC	No	No	Yes	Subject	Distant	Gender	Preverbal	-	-	-	No	-
Laurinavichyute et al. (2017) E2A	Russian	109	RC	No	No	Yes	Subject	Distant	Gender	Preverbal	-	-	-	cri-1,+2 (INH)	-
Laurinavichyute et al. (2017) E2B	Russian	112	RC	No	No	Yes	Subject	Distant	Gender	Postverbal	-	-	-	cri-3 (INH)	-
Pizarro-Guevara and Dillon (2021)	Tagalog	80	RC	No	No	Yes	Subject	Distant	Number	Postverbal	cri	-	cri	N/R	N/R
Yon (2019) E8	Korean	24	AC	No	No	Yes/No	Subject	Distant	Animacy	Preverbal	-	-	-	cri,+1 (INH)	-

Table A.10: A review of the studies on the real-time processing of local anaphors, grouped by the methodology employed. The review excludes studies that do not have structurally unavailable distractors (e.g., Dillon et al., 2014), or those with anaphors that are analyzed as exempt from standard binding constraints such as those in picture noun phrases (e.g., Runner et al., 2003). Unless stated as *recip(ocal)* under *Language*, all studies were on reflexives. *N* refers to the number of participants. The columns between *Embedding of target or distractor* and *Position of anaphor* detail experimental stimuli, and those after summarize empirical findings. *Clause-mate distractor?*, *Case matching distractor?* and *Matching feature of distractor* refer to match between the target and distractor. *Target match effects?* and columns including distractor match effects refer to differences between feature-matching vs. -mismatching conditions. For *Target match effects?*, *Bias towards target vs. distractor?*, and *Distractor match effects?*, ‘Yes’ means a significant effect, ‘No’ means a non-significant effect, ‘N/R’ means the test was not run or reported, and ‘-’ that the test does not apply. Statistical inferences follow main analyses originally reported as significant, and excludes exploratory analyses or marginal effects. Some studies are split by measure or stimuli, and thus appear on multiple lines. AC = adverbial clause, cri = critical region with anaphor, cri-/+ = region before or after the critical region as indexed with the following number, E = early reading measures in eye-tracking, FAC = facilitatory effects (i.e., speed-up with a feature-matching distractor), INH = inhibitory effects (i.e., slow-down with a feature-matching distractor), L = late reading measures in eye-tracking, NP = noun phrase, SC = sentential complement, PP = pre/post-positional phrase, RC = relative clause, WM = working memory.

Appendix B. Offline norming tasks for Experiments 1-3

Appendix B.1. Experiment 1: Antecedent selection task

Prior to the visual world study in Experiment 1, an offline antecedent selection task ($N_{\text{participant}} = 32$, $M_{\text{age}} = 22.8$) was conducted to verify the c-command constraint in the experimental stimuli. The task was conducted online on PCIbexFarm (Zehr and Schwarz, 2018). Each sentence was presented on the same screen along with a question asking about the referent(s) of the anaphor and four referent options, i.e., target, distractor, matrix subject and ‘others’. The order of the referent options was randomized across items. Participants were instructed to click on the referent(s) that answered the question. They were specifically instructed to click on all possible options, if more than one answer was possible. The experimental sentences were interspersed in a pseudo-randomized order among 36 fillers, including those probing the availability of multiple referents as correct answers. Each participant received 80 Turkish liras ($\sim \$4.5$ at the time of the study) for compensation.

Participants’ ($N = 31$; 1 excluded due to low accuracy) responses are summarized in Table B.11. Participants chose the target antecedent above 98% of the time across the experiment, confirming that the c-commanding, plural subject was a possible antecedent for *birbirleri*. The non-c-commanding distractor was overall chosen only around 14% of the time.

	Target	Distractor	Matrix Subject	Others
Recent Target, Distractor Match	97.8% (1.2)	15.5% (1.6)	2.2% (1.4)	4.8% (2.1)
Recent Target, Distractor Mismatch	100%	10.8% (1.8)	5.4% (3.2)	3.2% (1.6)
Distant Target, Distractor Match	97.8% (1)	19.9% (1.9)	4.3% (1.5)	5.6% (1.5)
Distant Target, Distractor Mismatch	98.9% (0.8)	11.3% (1.5)	5.8% (1.9)	5.9% (1.8)

Table B.11: Mean percentages of antecedent choices (and Cousineau-Morey corrected SEs) in the antecedent selection task for Experiment 1.

A mixed-effects logistic regression model was fit to the choice data for the distractor to systematically examine the distractor choices. The results revealed a main effect of MATCH, with more choices of a plural than a singular distractor ($z = 2.83$). A post-hoc comparison showed that there were more choices of a plural than a singular distractor when the target was distant ($z = 2.83$), but choices of a singular and plural distractor were comparable when the target was recent ($z = 1.76$). Overall, the task revealed some amount of distractor choices when it was plural and linearly proximate to the anaphor, although the overall rate of these responses is quite low.

Appendix B.2. Experiment 2: Norming task for the c-command relation of the IO

The indirect object (IO) of a ditransitive verb in Turkish may originate in a position that is higher or lower than the direct object (DO) (Öztürk, 2005). Whereas the IO c-commands the DO in its base position in the former case, it doesn't in the latter. To confirm that our stimuli had the IO c-command the anaphor in the DO position in its base position, we conducted a norming task with various ditransitive verbs. Twenty-four native speakers of Turkish ($M_{age} = 24.6$) participated in this task and received 100 Turkish liras ($\sim \$4$ at the time of the study) for compensation. Their task was to decide on the availability of IO>DO and DO>IO ordering of 16 ditransitive verbs when a bound-variable reading between a quantified IO and a lower, possessed DO is enforced as in *Ayşe asked every_i director about their_i cameraman* vs. *Ayşe asked about their_i cameraman to every_i director*. For twelve verbs, participants chose the IO>DO ordering over the DO>IO one in more than 70% of cases, suggesting that the IO c-commands the DO in the configurations formed with these verbs. Seven of these verbs that can be optionally used as a transitive or ditransitive verb were used as the embedded verb in the experimental stimuli of Experiments 2-3.

Appendix B.3. Experiment 2: Antecedent selection task

Prior to the visual world study in Experiment 2, an offline antecedent selection task ($N_{participant} = 42$, $M_{age} = 25.2$) was again conducted to verify the c-command constraint in the experimental stimuli. The experimental sentences were interspersed in a pseudo-randomized order among 42 fillers, including those probing the availability of multiple referents as well as those that required a dative-marked noun as the correct answer to ensure that participants were considering dative-marked nouns in the experimental sentences as options. The rest of the procedure was the same as in the previous antecedent selection task. Each participant received 100 Turkish liras ($\sim \$4$ at the time of the study) for compensation.

Participants' ($N = 39$; 3 excluded due to low accuracy) choices of antecedents are summarized in Table B.12. Results demonstrated that overall participants chose the embedded subject above 91% of the time. The IO was chosen around 35%, and the distractor was chosen around 27% of the time across the experiment.

A mixed-effects logistic regression model fit to the choice data for IO/Distractor in the 2x2 design, i.e., the four conditions with a plural subject, showed a main effect of the number of dative nouns, with more choices of plural than singular dative nouns ($z = 3.70$). A post-hoc comparison showed that plural IOs were chosen more often than singular ones ($z = 2.91$), but the choices for singular and plural distractors were comparable ($z = 1.74$). A t -test comparing the choices of the singular subject and

	Embedded Subject	IO/ Distractor	Matrix Subject	Others
Plural Subject, IO Match	93.2% (1.5)	33.8% (3.9)	2.1% (1)	3.4% (1.6)
Plural Subject, IO Mismatch	96.6% (1.3)	20.9% (3.1)	1.7% (0.9)	4.3% (1.2)
Plural Subject, Distractor Match	98.7% (1.2)	29.9% (4)	0.9% (0.7)	2.9% (1)
Plural Subject, Distractor Mismatch	97% (1.7)	23.9% (4.1)	0.4% (0.9)	3.8% (1.2)
Singular Subject, IO Match	71.8% (3.4)	49.6% (4.5)	4.7% (1.6)	11.1% (2.7)

Table B.12: Mean percentages of antecedent choices (and Cousineau-Morey corrected SEs) in the antecedent selection task for Experiment 2.

plural IO in the control condition showed significantly more choices of the subject ($t = 13.98$).

First, we found that plural IOs were more frequently selected as antecedents than singular IOs; confirming the availability of c-commanding, plural IOs as antecedents. However, singular IOs were still chosen approximately 20% of the time along with the plural antecedent chosen around 93% of the time. This suggests that *birbirleri* can, to some extent, be jointly bound by multiple c-commanding antecedents in its clause. Second, the overall selection rate of IOs was much smaller than that of subjects, suggesting that *birbirleri* is preferably bound by antecedents in the subject position. Third, in the control condition, although the plural IO was chosen more often than in the other conditions, the singular subject was still chosen more often than the IO despite its number mismatch, indicating that c-command and subjecthood strongly influenced antecedent selection. Interestingly, selections of the matrix subject and unspecified ‘other’ antecedents also increased in the control condition, possibly reflecting participants’ attempts to reconcile the plural requirement of *birbirleri* by considering additional antecedents beyond the binding domain.

Lastly, the overall selection rate of distractors was unexpectedly high, and feature-matching distractors were not reliably more available than mismatching ones. This may be because the argument of the postposition *göre* ‘according to’ may sometimes bind the reciprocal through a wide scope reading (see Experiment 2 for details). To test whether the distractor is less available when participants are limited to a single choice, we conducted a forced-choice task using the same stimuli (see below).

Appendix B.4. Experiment 2: Forced-choice antecedent selection task

To address the issue of high selection rate of distractors in the antecedent selection task (see above), we put the same materials from Experiment 2 into a forced choice antecedent selection task ($N_{\text{participant}} = 35$, $M_{\text{age}} = 27.5$). The list of experimental and filler items was the same as in the antecedent selection task above. The

procedure of this task was similar to the visual world study. That is, the participants were first given the question only, and then on a new screen the sentence along with the four referent options: embedded subject, IO/distractor, matrix subject and ‘others’. They were instructed to click on the most likely referent that answered the question. Each participant received 120 Turkish liras ($\sim$ \$4 at the time of the study) for compensation.

Participants’ ($N = 30$; 5 excluded due to low accuracy) choices of antecedents are summarized in Table B.13. Results demonstrated that overall participants chose the embedded subject above 85.1% of the time. The IO was chosen around 19.8%, and the distractor around 2.2% of the time across the experiment.

	Embedded Subject	IO/ Distractor	Matrix Subject	Others
Plural Subject, IO Match	80% (2.7)	17.8% (3)	0%	2.2% (1.1)
Plural Subject, IO Mismatch	91.6% (2.8)	7.8% (2.8)	0%	0.6% (1.1)
Plural Subject, Distractor Match	97.2% (1.8)	2.2% (1.6)	0%	0.6% (0.7)
Plural Subject, Distractor Mismatch	95.6% (2.4)	2.2% (2)	0%	2.2% (1.1)
Singular Subject, IO Match	61.1% (5.7)	33.9% (5.4)	0%	5% (1.9)

Table B.13: Mean percentages of antecedent choices (and Cousineau-Morey corrected SEs) in the forced-choice antecedent selection task for Experiment 2.

A mixed-effects logistic regression model fit to the choice data for IO/Distractor in the 2x2 design subject showed a main effect of c-command, with more choices of IO than distractor ($z = 4.54$). A post-hoc comparison showed that the plural IO was chosen more often than the singular IO ($z = 2.44$), but the choices for singular and plural distractors were comparable ($z = 0.11$). A t -test comparing the choices of the subject and IO in the control condition showed more choices of the subject ($t = 9.58$).

Overall, the forced-choice task showed that distractor selections were negligible when participants were limited to a single antecedent choice, in line with our speculation that the higher distractor selection rate above may stem from participants’ efforts to accommodate a wide-scope reading of the postposition *göre* ‘according to’ when multiple selections were allowed. However, in the control condition, singular c-commanding subjects were chosen more often than plural IOs, and in the Plural Subject with IO Mismatch condition, singular IOs were selected over plural subjects approximately 8% of the time; suggesting that c-command and subjecthood may play a more dominant role in antecedent selection than number alone.

Appendix B.5. Experiment 3: Antecedent selection task

Prior to the visual world study in Experiment 3, an offline antecedent selection task ($N_{\text{participant}} = 30$, $M_{\text{age}} = 25.7$) was again conducted to verify the c-command constraint the experimental stimuli. Twenty-four sets of experimental sentences were interspersed in a pseudo-randomized order among 46 fillers, including those probing the availability of multiple referents. The rest of the procedure was the same as in previous antecedent selection tasks. Each participant received 120 Turkish liras ($\sim \$4$ at the time of the study) for compensation.

Participants' ($N = 25$; 5 excluded due to low accuracy) choices of antecedents are in Table B.14. Results showed that the embedded subject was chosen above 95% of the time; the c-commanding IO was chosen around 46% of the time; and the distractor around 16% of the time across the experiment. These results confirmed that subjects and IOs can bind *birbirleri*, though IOs were again less available than the subject.

	Embedded Subject	IO/ Distractor	Matrix Subject	Others
IO, Dative	92% (1.6)	46% (4.6)	4.5% (0.8)	4% (1)
Distractor, Dative	99% (1.1)	12.5% (2.8)	4.5% (0.8)	3.9% (0.8)
Distractor, Genitive	95.5% (1.6)	20% (3.2)	6% (1.2)	6.5% (1.7)

Table B.14: Mean percentages of antecedent choices (and Cousineau-Morey corrected SEs) in the antecedent selection task for Experiment 3.

A mixed-effects logistic regression model fit to the choice data for IO/distractor revealed a main effect of c-command, with more choices of IO than distractor ($z = 5.17$). A comparison within the two distractor conditions showed that the genitive-marked distractor was chosen more often than the dative-marked distractor ($z = 2.45$). The increased choices of the genitive-marked distractors may reflect that participants occasionally considered these distractors as potential antecedents due to their case similarity to c-commanding subjects. Regardless of this effect, the choice data confirmed the unavailability of distractors in binding the reciprocal.

Appendix C. Raw gaze probabilities for the experimental sentence in Experiment 1

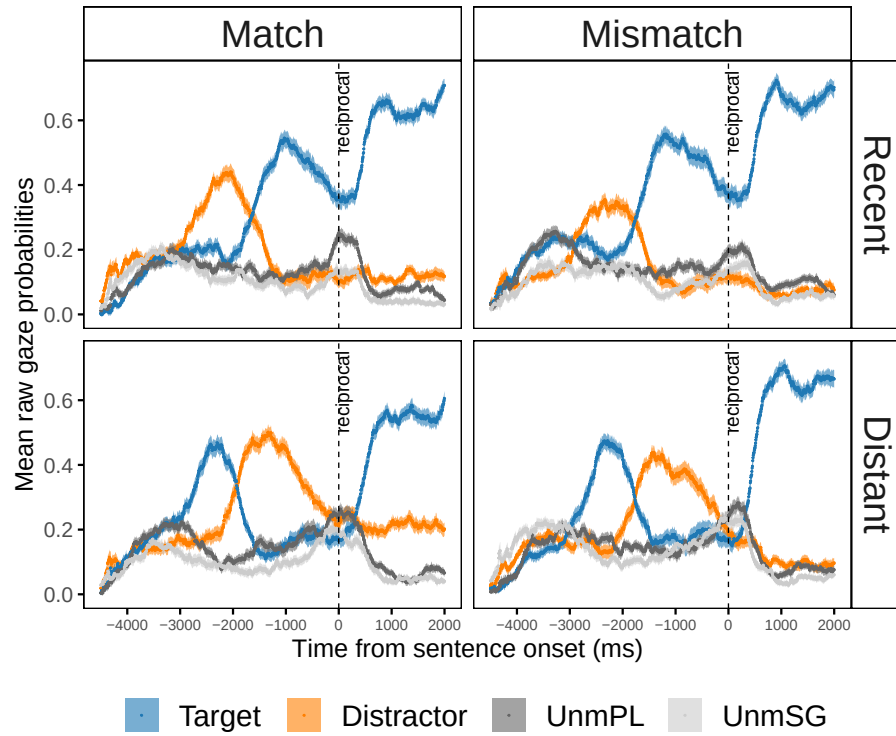


Figure C.12: Mean raw gaze probabilities towards the four interest areas in each condition in Experiment 1. Error bands show Cousineau-Morey corrected, within-participant SEs. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Probabilities are shown for the duration of the entire sentence, with dashed lines at time point 0 time-locked to the onset of the critical anaphor.

Appendix D. Visual and auditory stimuli in Experiments 1-3

All visual stimuli used in Experiments 1–3 were downloaded from Adobe Stock photos and were drawn by the same artist in a consistent style and format. Prior to the visual world studies, the stimuli were normed for codability. The participants ($N = 20$, $M_{age} = 22.7$) were asked to type the occupation of the character in each image in an online survey. Each participant received 80 Turkish liras ($\sim \$4.5$ at the time of the study) for compensation. Seventy-four of the pictures that were given the same label by at least 75% of the participants were used in the experiments.

The auditory stimuli in Experiments 1-3 were recorded using the GarageBand application on a Macbook Air by the first author. Each experimental sentence was recorded in three parts and then merged into a single audio file: (i) the matrix subject, (ii) the segment containing the target and distractor phrases (which varied by condition), and (iii) the final segment containing the adverb, anaphor, and two verbs. The same audio files were used for the first and third segments across all conditions of an item. To ensure that the critical anaphor onset was aligned across conditions, the three segments were merged using fixed onset points for each item. If the second segment was shorter in a condition, silent padding was added to its end so that the third segment began at a consistent time.

The three-part recording method introduced two prosodic boundaries, which presumably enhanced naturalness by inducing pauses at appropriate points. A pause following the matrix subject is expected in Turkish, as subjects are typically default topics and optionally followed by a pause (Göksel and Kerslake, 2005; Kamali, 2008). A pause before the adverb is also plausible, as the preceding segment contains three words, well within the typical prosodic phrase length observed in Turkish speakers who pause at every 4.2 words on average (range: 2.9–7.8) during read-aloud story tasks (according to Deniz’s (2014) analysis of Nash’s (1973) data).

Before data collection, the critical time points for the onsets of the reciprocal and embedded verb windows in each item set were established by the first author via examination of spectrograms in Praat.

Filler items were recorded as single segments but included two prosodic breaks in positions identical to those in the experimental sentences. Lead-in sentences and questions were recorded individually, as a single segment. To normalize for loudness, the average intensity of all recordings was set to 70.00 dB. The set of auditory files for the example stimuli in Experiment 1 (i.e., the stimuli in (4) and (5) in Section 3.1) can be found at the OSF repository.

Appendix E. Model formulas for the statistical analyses in Experiments 1-3

Appendix E.1. Model formulas in Experiment 1

Gaze probability analyses

- Reciprocal, crossed: Target vs. distractor $\sim$ Recency * Match + (1 + Recency + Match | participant) + (1 + Match + Match:Recency || item), optimizer = ‘bobyqa’
- Reciprocal, nested: Target vs. distractor $\sim$ Recency + Recent: Match + Distant: Match + (1 + Recency + Distant: Match | participant) + (1 + Recent: Match + Distant: Match || item)
- Embedded verb, crossed: Target vs. distractor $\sim$ Recency * Match + (1 + Recency + Match || participant) + (1 + Recency * Match || item)
- Embedded verb, nested: Target vs. distractor $\sim$ Recency + Recent: Match + Distant: Match + (1 + Recency + Distant: Match || participant) + (1 + Recency + Distant: Match | item), optimizer = ‘bobyqa’

Analyses on new fixations towards target

- Reciprocal, crossed: Target $\sim$ Recency * Match + (1 + Recency * Match || participant) + (0 + Recency | item)
- Reciprocal, nested: Target $\sim$ Recency + Recent: Match + Distant: Match + (1 + Recency + Recent: Match + Distant: Match || participant) + (0 + Recency + Recent: Match | item), optimizer = ‘bobyqa’
- Embedded verb, crossed: Target $\sim$ Recency * Match + (0 + Match | participant) + (1 + Recency || item)
- Embedded verb, nested: Target $\sim$ Recency + Recent: Match + Distant: Match + (1 + Recent: Match + Distant: Match || participant) + (1 + Recency || item)

Analyses on new fixations towards distractor

- Reciprocal, crossed: Distractor $\sim$ Recency * Match + (1 + Match | participant) + (1 + Match + Recency:Match || item)

- Reciprocal, nested: Distractor $\sim$ Recency + Recent: Match + Distant: Match + (1 | participant) + (1 | item)
- Embedded verb, crossed: Distractor $\sim$ Recency * Match + (1 + Match || participant) + (1 + Match + Recency:Match || item)
- Embedded verb, nested: Distractor $\sim$ Recency + Recent: Match + Distant: Match + (1 | participant) + (1 + Recent: Match + Distant: Match || item)

Appendix E.2. Model formulas in Experiment 2

Gaze probability analyses

- Reciprocal, crossed: Subject vs. IO/distractor $\sim$ C-command * Match + (1 + C-command | participant) + (1 + C-command || item)
- Reciprocal, nested: Subject vs. IO/distractor $\sim$ C-command + IO: Match + Distractor: Match + (1 + C-command | participant) + (1 + C-command || item)
- Embedded verb, crossed: Subject vs. IO/distractor $\sim$ C-command * Match + (1 | participant) + (1 + Match || item)
- Embedded verb, nested: Subject vs. IO/distractor $\sim$ C-command + IO: Match + Distractor: Match + (1 | participant) + (1 + IO: Match | item)

Analyses on new fixations towards subject

- Reciprocal, crossed: Subject $\sim$ C-command * Match + (1 + C-command + Match || participant) + (1 + Match + C-command:Match | item)
- Reciprocal, nested: Subject $\sim$ C-command + IO: Match + Distractor: Match + (1 + C-command || participant) + (0 + IO: Match + Distractor: Match | item)
- Embedded verb, crossed: Subject $\sim$ C-command * Match + (1 + C-command * Match || participant) + (0 + C-command:Match | item)
- Embedded verb, nested: Subject $\sim$ C-command + IO: Match + Distractor: Match + (1 + C-command + IO: Match | participant)

Analyses on new fixations towards IO/distractor

- Reciprocal, crossed: $\text{IO/distractor} \sim \text{C-command} * \text{Match} + (1 + \text{C-command} \parallel \text{participant}) + (0 + \text{C-command} + \text{C-command:Match} \parallel \text{item})$
- Reciprocal, nested: $\text{IO/distractor} \sim \text{C-command} + \text{IO: Match} + \text{Distractor: Match} + (1 + \text{C-command} \parallel \text{participant}) + (0 + \text{C-command} \parallel \text{item})$
- Embedded verb, crossed: $\text{IO/distractor} \sim \text{C-command} * \text{Match} + (1 + \text{C-command} \mid \text{participant}) + (1 + \text{Match} + \text{C-command:Match} \parallel \text{item})$
- Embedded verb, nested: $\text{IO/distractor} \sim \text{C-command} + \text{IO: Match} + \text{Distractor: Match} + (1 + \text{C-command} \mid \text{participant}) + (1 + \text{IO: Match} \parallel \text{item})$

Appendix E.3. Model formulas in Experiment 3

Gaze probability analyses

- Reciprocal: Subject vs. $\text{IO/distractor} \sim \text{C-command} + \text{Case} + (1 + \text{C-command} + \text{Case} \mid \text{participant}) + (1 + \text{Case} \parallel \text{item})$, optimizer = ‘bobyqa’
- Embedded verb: Subject vs. $\text{IO/distractor} \sim \text{C-command} + \text{Case} + (1 + \text{C-command} + \text{Case} \mid \text{participant}) + (1 + \text{Case} \parallel \text{item})$

Analyses on new fixations towards subject

- Reciprocal: $\text{Subject} \sim \text{C-command} + \text{Case} + (1 \mid \text{participant}) + (1 \mid \text{item})$
- Embedded verb: $\text{Subject} \sim \text{C-command} + \text{Case} + (1 + \text{C-command} \parallel \text{participant}) + (1 + \text{C-command} \parallel \text{item})$

Analyses on new fixations towards IO/distractor

- Reciprocal: $\text{IO/distractor} \sim \text{C-command} + \text{Case} + (1 + \text{C-command} \mid \text{participant}) + (1 \mid \text{item})$
- Embedded verb: $\text{IO/distractor} \sim \text{C-command} + \text{Case} + (0 + \text{C-command} \mid \text{participant}) + (1 \mid \text{item})$

Appendix F. Raw gaze probabilities for the experimental sentence in Experiment 2

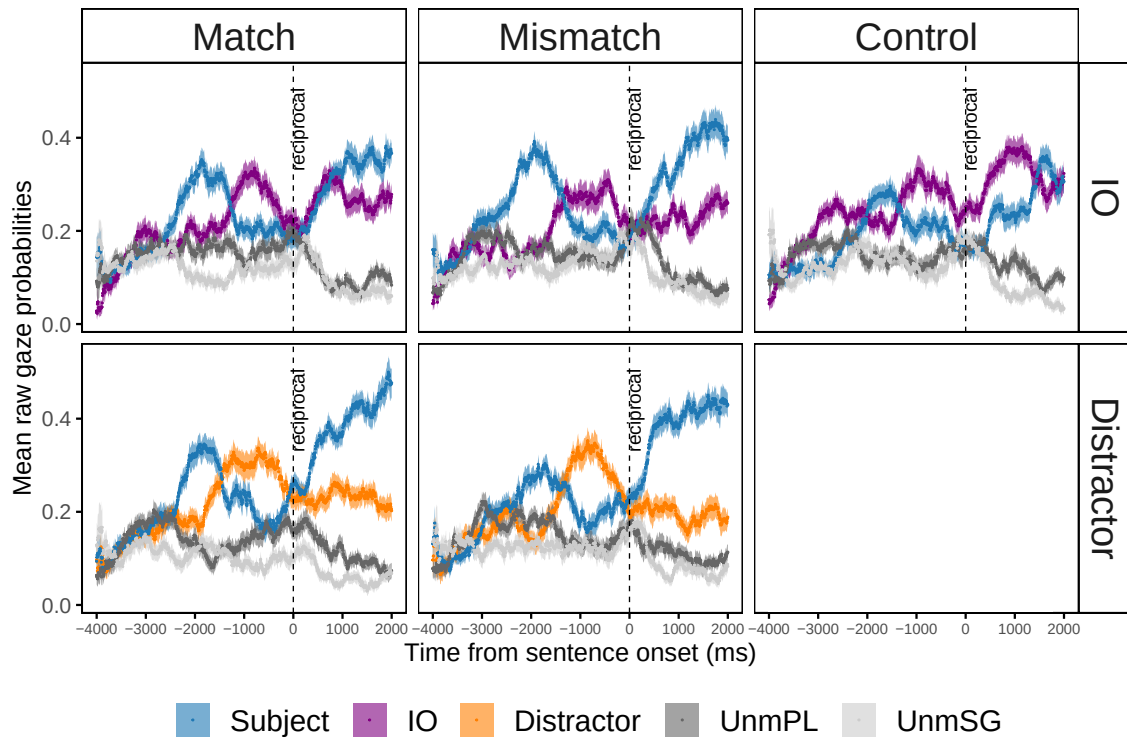


Figure F.13: Mean raw gaze probabilities towards the four interest areas in each condition in Experiment 3. Error bands show Cousineau-Morey corrected, within-participant SEs. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Probabilities are shown for the duration of the entire sentence, with dashed lines at time point 0 time-locked to the onset of the critical anaphor.

Appendix G. Raw gaze probabilities for the experimental sentence in Experiment 3

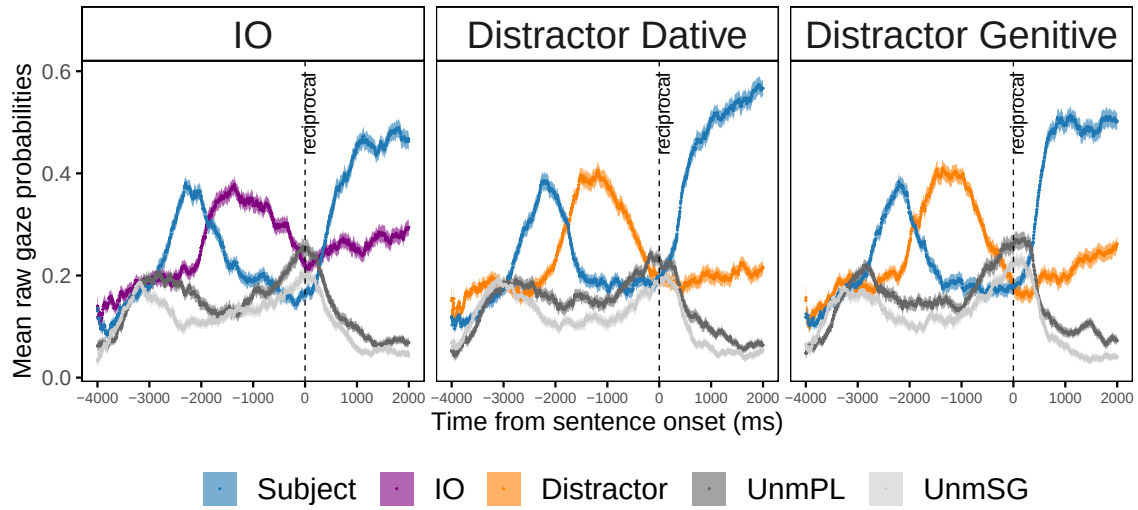


Figure G.14: Mean raw gaze probabilities towards the four interest areas in each condition in Experiment 3. Error bands show Cousineau-Morey corrected, within-participant SEs. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Probabilities are shown for the duration of the entire sentence, with dashed lines at time point 0 time-locked to the onset of the critical anaphor.

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